

Insider Trade Disclosure, Market Efficiency, and Liquidity

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Abstract

Is a more transparent market also more efficient and liquid? We address this question by analyzing the impact of mandatory ex-post disclosure of corporate insider trades in a dynamic model of strategic risk averse insider trading. We show that trade disclosure reduces informational efficiency of prices, may cause the market to be less liquid, and may even increase insiders' expected utility. In a more transparent market, the informed trader uses a less aggressive and noisy trading strategy in order to prevent the market maker from perfectly inferring the private information from public records, and to maintain her informational advantage over time. Our results then question the effectiveness of disclosure regulations.

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1 Introduction

Market transparency has been for many years at the heart of policy debates concerning the design of securities markets. However, there is no agreement among financial market regulators on what should be the optimal degree of transparency. In the United States, the Securities and Exchange Commission's (SEC) view is straightforward: market transparency "plays a fundamental role in the fairness and efficiency of the secondary market, [...] improves the price discovery, fairness, competitiveness and attractiveness of U.S. market" (SEC, 1995). Similarly, the International Organization of Securities Commissions (IOSCO) states that "market transparency – in essence, the widespread availability of information relating to current opportunities to trade and recently completed trades – is generally regarded as central to both the fairness and efficiency of a market, and in particular to its liquidity and quality of price-formation" (IOSCO, 2001). In contrast, in the United Kingdom, the Securities and Investment Board (SIB) has argued that there is "a tradeoff between liquidity and trade transparency" and consequently "transparency should be restricted if this is necessary to assure adequate liquidity" (SIB, 1994).

This paper examines ex-post disclosure of corporate insider trades, as an important aspect of market transparency. In particular, we provide new insights on how strategic insider trading is affected by disclosure requirements in a dynamic setting. We find that mandatory trade disclosure induces corporate insiders to trade less aggressively and occasionally in a direction opposite to their information. Because of this effect, we show that trade disclosure can slow down information revelation and reduce market liquidity when insiders are risk averse. This is in contrast to previous results in the literature, obtained under the assumption of risk neutrality.

According to Section 16(a) of the Securities Exchange Act of 1934, insider traders (officers and directors) must report to the SEC transactions in equity securities directly with the related issuer within ten days following the end of the month in which the trade had occurred. The rationale for this securities regulation, as recognized by the authority, is to make private information available to all market participants more rapidly, thus increasing price efficiency and market liquidity:

Section 16(a) is likely to provide significant benefits by making information concerning insiders' transactions in issuer equity securities publicly available substantially sooner than it was before. Making this information available to all investors on a more timely basis should increase market transparency, which will likely enhance market efficiency and liquidity.

— U.S. Securities and Exchange Commission, File No. S7-31-02.

Recently, after the introduction of the Sarbanes-Oxley Act (August 2002), the US financial market regulator tightened up this regulation by requiring insiders to report their trades not later than two business days following the transaction. This drastic change highlights the intention of the

regulator to reduce the degree of asymmetric information in the market.

A similar rule is enforced in the U.K., where corporate insiders must inform their company as soon as possible and no later than the fifth business day after a transaction for their own account or on behalf of their spouses and children. In turn, a company must inform the London Stock Exchange (LSE) without delay and no later than the end of the business day following receipt of the information.¹ Therefore, an improvement in market transparency through the disclosure of insider trades should translate into a higher liquidity and efficiency of the market. In this paper we present a theoretical model to study whether and to which extent this is true.

We consider a rational expectation trading model in the spirit of Kyle (1985) in which a risk averse corporate insider has long-lived private information and is subject to a mandatory ex-post trade disclosure regulation. We model the regulation by requiring the informed agent to disclose ex-post her last trade. By comparing equilibrium outcomes for the same economy with and without regulation (hereafter referred as transparent and opaque market, respectively) we analyze the impact of an increase in transparency on the efficiency and liquidity of the market and on the profitability of informed trading.

Our main result is that transparency reduces informational efficiency of prices and can cause the market to be less liquid. The analysis of the following two scenarios should clarify the intuition behind this somewhat surprising result. In an opaque market (i.e. without disclosure regulation), as showed by Holden and Subrahmanyam (1994) and Baruch (2002), a risk averse informed trader chooses to exploit her informational advantage very rapidly to protect herself against future price risk imposed by liquidity traders. This aggressive trading behavior affects positively both market efficiency and liquidity. Indeed, almost all the private information is revealed to the market at the very early stages of the trading rounds, thus strongly reducing the adverse selection problem faced by the market maker. This implies a very liquid market. In a transparent market (i.e. with disclosure regulation), instead, we find an equilibrium in which the aggressiveness of the informed trader is severely reduced. The introduction of the disclosure requirement creates a tradeoff in the informed agent's strategy between future price risk and the revelation of private information through disclosure. In equilibrium the concern for the latter prevails.

There are indeed two opposite effects associated with the enforcement of such securities regulation. The positive direct effect is represented by the flow of information disclosed by the insider at the end of any trading round that clearly decreases the uncertainty caused by liquidity traders' order flow. This by itself reduces the adverse selection problem of the market maker. However, the indirect and negative effect, due to the change in the insider's trading strategy, intensifies the degree of asymmetric information. In this paper we show that when the informed agent is risk

¹This implies that information about an insider transaction can reach the market as late as 6 days after the transaction. However, in practice, this information is disclosed faster: as documented by Fidrmuc et al. (2006) the announcement day for most of the directors' dealings (85% in their sample) coincides with the transaction day or is the following day.

averse the indirect effect exist, and most importantly it dominates.

Consistently with Huddart et al. (2001) and Zhang (2004), the insider plays a mixed strategy by adding a random component to her order flow. This prevents the market maker from inferring perfectly the private information from public records, and allows the insider to maintain an informational advantage over time. A lower market efficiency is caused by the fact that private information is now slowly incorporated into prices. One interpretation of this result is that corporate trade disclosure can be seen as the institutional means that forces a risk averse insider to behave as if she were risk neutral. In order to sustain an equilibrium in mixed strategy, trading costs set by the market maker must be constant over time, and this, indeed, corresponds to the well-known condition that characterizes a risk neutral trading behavior. An interesting feature of the model is that the transparent market equilibrium is independent of the level of risk aversion when trading is continuous.

Regarding the profitability of insider trading, we show that when the private information owned by the corporate insider is sufficiently unexpected – that is when the degree of asymmetric information is sufficiently high – the introduction of a trade disclosure regulation increases the insider's expected utility. In particular, we show that her expected utility can be easily interpreted a function of the positive direct and negative indirect effects associated with the enforcement of the regulation, where the relative weight of the negative effect (and hence positive for the insider profitability) is given by how unexpected her private information is.

More in details, our results can be summarized as follows: when ex-post insider trade disclosure is imposed (i) the insider maintains an informational advantage over the entire trading period despite the disclosure; (ii) market is less efficient at any trading round; (iii) the difference in market inefficiency, in terms of how much of the private information is not incorporated into prices, even widens if risk aversion/the variance of the liquidation value of the risky asset/the volatility of liquidity trading increase; (iv) trading prices have constant volatility over time and information is incorporated into prices at a constant rate; (v) even when the insider is risk neutral market efficiency does not improve if trading is continuous; (vi) market liquidity is constant over time; (vii) aggregate execution cost increases if the insider is sufficiently risk averse; (viii) the insider's ex-ante expected utility, conditional on her private information, increases if this information is sufficiently unexpected. These results thus question the effectiveness of such securities regulation.

This paper relates to the literature on market transparency, in which the main focus is the tradeoff between (pre- and post-trade) transparency and liquidity. Analytical results (Madhavan, 1995; Pagano and Röell, 1996; Naik et al., 1999; Frutos and Manzano, 2002), laboratory experiments (Bloomfield and O'Hara, 1999; Flood et al., 1999) and natural experiments (Madhavan et al., 2005) have shown that in a dealer market transparency improves informational efficiency but causes opening spreads to widen.² Verrecchia (2001) and more recently Leuz and Wysocki (2008)

²Flood et al. (1999), instead, find that quote transparency reduces both opening spreads and market efficiency.

provide comprehensive reviews on the economic consequences of financial reporting and disclosure regulation, and emphasize the importance of studying the effects of disclosure in dynamic settings. Dierker and Subrahmanyam (2013) considers a dynamic noisy rational expectation model where managers can time corporate disclosures, and finds that firms may maximize the informational efficiency of their stock prices by postponing disclosure.

This paper also relates to the literature on strategic insider trading with mandated disclosure. Fishman and Hagerty (1995) consider a two period model in which an insider may become informed with a certain probability. When the insider becomes informed, she never manipulates the market, while she might do it (imitating an informed insider with good news) if uninformed. John and Narayanan (1997) extend their model by introducing asymmetry in the probability of receiving a good or a bad signal. In this setting also an informed insider may manipulate the market. In both models manipulation is driven by an uninformed insider's attempt to pool with an informed insider and never occurs in equilibrium with good and bad news simultaneously. The most related article to this paper is Huddart et al. (2001). They extend the discrete time model by Kyle (1985) with risk neutral insider trading by adding the trade disclosure constraint. They find that an equilibrium in which the insider adds a noise component to her trading strategy (*dissimulation strategy*) exists and that the regulation accelerates price discovery and increases market depth. In this paper we show that this result is driven by the restrictive assumption of risk neutrality. Cao and Ma (1999) introduce imperfect competition among insiders. In their model market efficiency is unambiguously higher with disclosure, while market liquidity may be lower when insiders' signal are negatively correlated.³

We contribute to the existing literature on market transparency and strategic insider trading by (i) examining a different aspect of transparency (the requirement for corporate insiders to ex-post disclose their trades) which is of primary importance for the design of securities markets; (ii) obtaining close-form solution for the transparent and opaque market equilibrium with risk aversion; (iii) showing the detrimental effect of transparency on market efficiency and liquidity.

Consistent with our *mixed strategy equilibrium*, empirical studies on legal insider trading show that insiders place both informed and uninformed trades, and that on average the information content is small. Lakonishok and Lee (2001) provides event-study evidence of statistically but not economically significant market reaction around US legal insider purchases. Fidrmuc et al. (2006) reports abnormal returns of higher magnitude for the UK. However, abnormal returns could be a noisy proxy for insider information and the possible endogenous relation between abnormal returns and insider trading may lead to inconsistent results. To overcome these problems, Aktas et al. (2008) measure the contribution of insider trades to market efficiency by estimating the correlation between returns and the relative order imbalance, a methodology recently introduced by Chordia et al. (2005). Within this setting they find that insiders contribute significantly to faster

³Cao and Ma (1999) adopt the same signals structure as in Foster and Viswanathan (1996) and use as a benchmark to their continuous time extension the model without trade disclosure by Back et al. (2000).

price discovery. Degryse et al. (2009) analyzes the information content of insider trades across different regulatory regimes. Using de-aggregated data for Dutch listed firms, the authors find that the implementation of the Market Abuse Directive (European Union Directive 2003/6/EC), which makes the reporting requirements harsher, reduces the information content of sales by top executive. This finding seems in line with our results.

The remainder of the paper is organized as follow. Section 2 introduces the features of the model. An opaque market equilibrium is derived in Section 3, whereas in Section 4 we solve for a transparent market equilibrium. In Section 5 we compare and discuss the properties of the equilibria, and assess the effectiveness of the disclosure regulation. In Section 6 we describe some findings concerning the insider ex-ante expected profits. Section 7 outlines the policy implications of our findings and concludes. In the Appendix we relegate all the proofs and the derivation of a transparent market equilibrium in discrete-time.

2 Model

Our model is based on Kyle (1985). Consider a single risky asset traded in continuous auctions by an informed trader (*the corporate insider*) and a number of liquidity traders. We assume that the sequence of auction dates $\langle t \rangle$ partitions the time interval $[0, 1]$. At any auction a market maker observes the total order flow and fixes the price at which all orders are filled. The corporate insider is risk averse, whereas the market maker is risk neutral.⁴

Let us denote with v the *ex-post* liquidation value of the risky asset, which is assumed to be normally distributed, $v \sim \mathcal{N}(p_0, \Sigma_0)$. The informed trader has perfect information in the sense she observes at $t = 0$ the liquidation value v .⁵ Let $dx(t)$ be the order placed by the monopolistic insider trader at time t . The aggregate order by liquidity traders, who trade for liquidity reasons, is denoted by $du(t)$, and we assume it follows an arithmetic Brownian motion,

$$du(t) = \sigma_u dB^u(t) \tag{1}$$

for some constant σ_u , independent of v .⁶ Furthermore, let $dy(t)$ be the total order flow given by the sum of the order flows place by the insider trader and the liquidity traders: $dy(t) = dx(t) + du(t)$.

We denote with $W(0)$ the initial wealth of the informed trader and with $W(t)$ the wealth at

⁴As argued by Holden and Subrahmanyam (1994), market making is typically performed by large financial institutions, which have large capacity to bear risk. Therefore, their behavior can be modeled by assuming risk neutrality.

⁵The assumption of perfectly informed monopolistic insider seems the most reasonable given the specific type of regulation we analyze: corporate insiders subject to this regulation are very likely to have perfect information and to be able to collude with each other, thus behaving as single informed agent.

⁶Back (1992), Back and Pedersen (1998), and Baruch (2002) consider a model in which the instantaneous variance σ_u deterministically changes over time. More recently, Collin-Dufresne and Fos (2012) considers the case of stochastic noise trading volatility. To keep our formulation tractable, we focus our analysis on the case of constant variance.

time t . The objective of the insider is to maximize the expected utility over terminal wealth $W(1)$. Preferences are described by exponential utility, with coefficient of absolute risk aversion equal to A :

$$U(W(1)) = -\exp\{-AW(1)\} \quad (2)$$

At time t the market maker sets the market price $p(t)$ by a competitive process, such that the price equals the expected value of v conditional on all public information available at that auction.⁷ The monopolistic insider instead chooses a trading strategy that maximizes her expected utility conditional on all public and private information. Equilibrium is therefore defined as follows:⁸

Definition 1. *A linear equilibrium of the continuous-time trading game is defined by the following conditions:*

- i. Given the linear pricing rule $p(t)$, the linear trading strategy $dx(t)$ maximizes*

$$\mathbb{E}[-\exp\{-AW(1)\}] \quad (3)$$

- ii. Given the linear trading strategy $dx(t)$, the pricing rule is competitive:*

$$p(t) = \mathbb{E}_t^m(v) \quad (4)$$

where the expectation is taken conditional on the market maker information set at time t .

3 Opaque Market Equilibrium

In this Section we characterize the market equilibrium in a setting with continuous trading and no mandatory disclosure: this is the benchmark that allows us to assess the effectiveness of the securities regulation. A more general version of this model, in which different elastic liquidity demand functions are considered, is studied in Baruch (2002). The next theorem shows how to construct a linear equilibrium.

Theorem 1. *There exists a recursive linear equilibrium in which the constants $\tilde{\lambda}(t)$, $\tilde{\beta}(t)$, $\tilde{\alpha}(t)$,*

⁷This is equivalent to assuming Bertrand competition among several market makers, implying that the profits of the market maker(s) are driven to zero.

⁸Since the aim of this paper is far from (technically) generalizing the Kyle (1985) model, we consider throughout the paper a locally linear choice space (space in which linear pricing rules and linear trading strategies are defined) in which doubling strategies are ruled out. We refer to Back (1992) for further details.

$\tilde{\delta}(t)$, and $\tilde{\Sigma}(t)$ characterize the following:

$$dx(t) = \tilde{\beta}(t)M(t)dt \quad (5)$$

$$dp(t) = \tilde{\lambda}(t)dy(t) \quad (6)$$

$$\tilde{\Sigma}(t) = \mathbb{V}\text{ar}[v|\mathcal{F}(t)] \quad (7)$$

$$J(M, t) = -\exp\{-A[W(t) + V(M, t)]\} \quad (8)$$

$$V(M, t) = \tilde{\alpha}(t)[M(t)]^2 + \tilde{\delta}(t) \quad (9)$$

for every $t \in [0, 1]$, where $M(t) \equiv v - p(t)$. The constants $\tilde{\beta}(t)$, $\tilde{\lambda}(t)$, $\tilde{\alpha}(t)$, $\tilde{\delta}(t)$, $\tilde{\Sigma}(t)$ are given by

$$\tilde{\Sigma}(t) = \frac{\left[2 + \left(A^2\Sigma(0)\sigma_u^2 - \sqrt{A^2\Sigma(0)\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)}\right)t\right]\Sigma(0)(1-t)}{2 + 2A^2\Sigma(0)\sigma_u^2(1-t)t} \quad (10)$$

$$\tilde{\lambda}(t) = \left[\frac{\sigma_u\sqrt{4 + A^2\Sigma(0)\sigma_u^2}}{2\sqrt{\Sigma(0)}} + A\sigma_u^2\left(t - \frac{1}{2}\right)\right]^{-1} \quad (11)$$

$$\tilde{\beta}(t) = \frac{2\sigma_u}{\left[\sqrt{\Sigma(0)(4 + A^2\Sigma(0)\sigma_u^2)} - A\Sigma(0)\sigma_u\right](1-t)} \quad (12)$$

$$\tilde{\alpha}(t) = \frac{1}{2\tilde{\lambda}(t)} \quad (13)$$

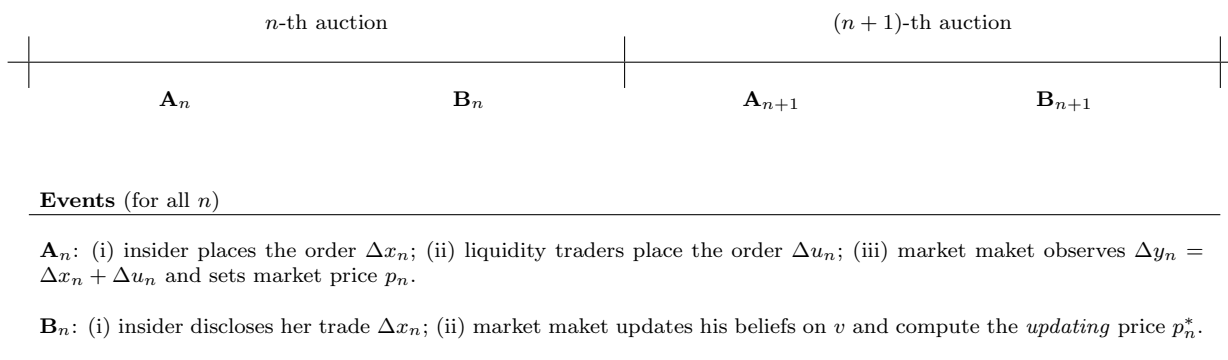
$$\tilde{\delta}(t) = \frac{\sigma_u^2}{2} \int_t^1 \tilde{\lambda}(s)ds \quad (14)$$

Proof. See the Appendix. □

The linear strategy of the corporate insider is characterized by $\tilde{\beta}(t)$, the one of the market maker by $\tilde{\lambda}(t)$. $\tilde{\Sigma}(t)$ denotes the conditional variance of prices, while $\tilde{\alpha}(t)$ and $\tilde{\delta}(t)$ characterize the value function of the insider.

A risk averse informed agent faces the following trade-off: on the one hand she would like to use all the private information immediately because concerned about future price risk induced by liquidity traders, on the other hand she would like to concentrate all her trade at the time in which the trading cost, $\tilde{\lambda}(t)$ is minimized. In equilibrium the optimal balance of these two components is determined by the decreasing dynamic of the trading cost. At the early stages of trading the insider behaves more aggressively, yet maintaining an informational advantage over time that allows her to trade till the last auction. At the end of the trading period, $t = 1$, all the private information is revealed through prices, $\tilde{\Sigma}(1) = 0$. When the corporate insider is risk neutral the concern for future price risk disappears, and only a constant trading cost can be part of an equilibrium. We refer to Baruch (2002) for further discussions and comparative statics.

FIGURE 1. DISCLOSURE REGULATION IN DISCRETE-TIME: THE SEQUENCE OF EVENTS



4 Transparent Market Equilibrium

In this Section we characterize the market equilibrium when mandatory trade disclosure is enforced. To easily communicate the intuition of how such regulation works in a dynamic model of strategic trading, we prefer to introduce and describe it in the context of a more intuitive discrete-time setting – in which we can make use of the time between two subsequent auctions – before moving to a more elegant and analytically tractable continuous-time framework.

Disclosure regulation in discrete-time. Mandatory trade disclosure is modeled by imposing the informed trader to reveal at the end of any auction the amount of trade she placed at the beginning of the auction. Hence the insider's past trades become public information. Therefore, the market maker, after having set the price at the n -th auction, but before setting the market price of the subsequent one, observes the order placed by the informed trader at the n -th auction. Thus, he adjusts his belief of the asset value using this new piece of information reported by the insider. We define p^* as the *updating* price, which is not a market price but rather the price that the market maker would have set for the n -th auction if he had known the insider's order at the beginning of the period. This *updating* price is computed by the market maker in order to determine the market price p of the next auction. The sequence of events in Figure 1 makes this clear. We leave in the Appendix the derivation of the transparent market equilibrium in the discrete-time setting, and we show at the end of the current Section that as the interval between auctions in the discrete-time model becomes uniformly small, the sequential auction equilibrium with mandatory trade disclosure converges to the equilibrium in continuous-time.

Disclosure regulation in continuous-time. The same intuition holds for the continuous-time setting, in which, instead, the insider simultaneously reveals at each instant t the trade made a

dt before and places the new order (i.e., the events $\mathbf{B}_n$ and $\mathbf{A}_{n+1}$ in Figure 1 happen at the same time). The main feature of the existing ex-post disclosure regulations is that the insider maintains an informational advantage between the trading and the disclosing periods. We keep this essential features in our continuous-time framework by assuming that the market maker, although continuously observing the insider's disclosed trades, uses the total order flow $dy(t)$ to determine the market price $p(t)$. Therefore, our continuous-time formulation should be interpreted as the limit of the discrete-time framework. We define the *updating* price, $p^*(t)$, which is not a market price but rather the price that the market maker would have set in the previous auction (a dt before) if he had been able to observe the insider's order. The sequence of updating prices will be used by the market maker as a "base" to determine the sequence of market prices.

Remark 1. *Consider a more general formulation in which the corporate insider trades and discloses continuously, but the disclosure of the trade at time t , $dx(t)$, occurs with a discrete lag Δ (according to the current U.S. regulation Δ would correspond to two business days). That is, at time t the insider places the trade $dx(t)$ and discloses $dx(t - \Delta)$. Our continuous-time formulation can be considered as the limit of this trading game when Δ becomes uniformly small. Therefore, in this paper we focus on the most extreme ex-post disclosure regulation with a disclosure delay of dt .*

As argued by Huddart et al. (2001), trading and pricing strategies characterizing a linear equilibrium in an opaque market can not be part of an equilibrium in a transparent market. To see this, suppose the contrary holds. Suppose the market maker conjectures that insider's demand is just a linear function of the liquidation value, v , of the risky asset. Then, after the first disclosure, the market maker would be able to invert the insider's trading strategy, thus perfectly inferring the liquidation value. So, no private information would be left to the insider. As a consequence, the market maker would set the price in the next auction equal to v by choosing zero sensitivity to the order flow in his pricing function (infinite market depth). Behaving rationally, the insider would anticipate this and have an incentive to deviate from her trading strategy in order to induce mispricing and make unbounded profits. Therefore, no invertible trading strategy can be part of an equilibrium with trade disclosure (*no equilibrium in pure strategy*).

However, an equilibrium in mixed strategy does exist. At any auction the insider's trading strategy is determined by two components: (i) a private information-based linear component, and (ii) a plain noise component. The next theorem presents the closed-form solution for the linear equilibrium of the trading game in the continuous-time framework.

Theorem 2. *There exists a recursive linear equilibrium in which the constants $\lambda(t)$, $\gamma(t)$, $\beta(t)$, $\alpha(t)$, $\delta(t)$, $\sigma_z(t)$, and $\Sigma(t)$ characterize the following:*

$$dx(t) = \beta(t)M(t)dt + dz(t) \quad (15)$$

$$dz(t) = \sigma_z(t)dB^z(t) \quad \text{with} \quad d\langle B^z, B^u \rangle_t = 0 \quad (16)$$

$$dp(t) = p^*(t) - p(t) + \lambda(t)dy(t) \quad (17)$$

$$dp^*(t) = \gamma(t)dx(t) \quad (18)$$

$$\Sigma(t) = \text{Var}[v|\mathcal{F}'(t)] \quad (19)$$

$$J(M, t) = -\exp\{-A[W(t) + V(M, t)]\} \quad (20)$$

$$V(M, t) = \alpha(t)[M(t)]^2 + \delta(t) \quad (21)$$

for every $t \in [0, 1]$, where $M(t) \equiv v - p^*(t)$. Given the initial condition $p^*(0) = p(0)$, the constants $\beta(t)$, $\lambda(t)$, $\gamma(t)$, $\alpha(t)$, $\delta(t)$, $\sigma_z(t)$, $\Sigma(t)$ are given by

$$\Sigma(t) = (1 - t)\Sigma(0) \quad (22)$$

$$\gamma(t) = \hat{\gamma} = (\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (23)$$

$$\lambda(t) = \frac{1}{2}\hat{\gamma} = \frac{1}{2}(\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (24)$$

$$\beta(t) = \hat{\gamma}^{-1}/(1 - t) = \sigma_u\Sigma(0)^{-\frac{1}{2}}/(1 - t) \quad (25)$$

$$\alpha(t) = \frac{1}{2}\hat{\gamma}^{-1} = \frac{1}{2}(\sigma_u^2/\Sigma(0))^{\frac{1}{2}} \quad (26)$$

$$\delta(t) = 0 \quad (27)$$

$$\sigma_z(t) = \sigma_u, \quad t \in (0, 1) \quad (28)$$

with the boundary condition $\sigma_z(1) = 0$.

Proof. See the Appendix. □

The linear strategy of the corporate insider is now characterized by $\beta(t)$ and $\sigma_z(t)$, the one of the market maker by $\lambda(t)$. $\gamma(t)$ represents the updating trading cost that the market maker can compute after any trade disclosure. $\Sigma(t)$ denotes the conditional variance of prices, and $\alpha(t)$ and $\delta(t)$ characterize the value function of the insider.

Since no invertible trading strategy can be part of an equilibrium, the informed agent adds a random component to her market order: at each trading round she chooses the optimal variance of such noise, $\sigma_z(t)$. Equation (28) states that in equilibrium this variance must be equal to the instantaneous volatility of liquidity trading, σ_u , except for the last trading round in which no

dissimulation component is needed. As a consequence, trading costs $\lambda(t)$ must be constant over time in order to make the insider indifferent among all the possible values of his order flow. Private information is incorporated into prices gradually and market prices have constant volatility over time. A somehow surprising feature of this equilibrium is that it is independent of the level of risk aversion. This and other properties are discussed in detail in the next Section.

A convergence result. In the Appendix we derive a transparency market equilibrium in discrete-time. We study now how the equilibrium properties of the sequential trading model are related to the properties of the continuous trading model when the interval between auctions become smaller and smaller. A convergence result shows that as the interval between auctions in the discrete-time model becomes uniformly small, the sequential auction equilibrium with mandatory trade disclosure converges to our continuous auction equilibrium.

Proposition 1. *Holding $\Sigma(0) = \Sigma_0$ and σ_u^2 constant, and given the convention $x(t) = x_n$ for all $t \in [t_{n-1}, t_n)$, consider a sequence of sequential equilibria such that $\Delta t \rightarrow 0$. Then the values $\Sigma(t), \gamma(t), \lambda(t), \beta(t), \alpha(t), \delta(t)$, and $\sigma_z(t)$ characterized in Theorem 3 converge to the corresponding value in the continuous auction equilibrium obtained in Theorem 2.*

Proof. See the Appendix. □

5 Effectiveness of Transparency Regulation

In the introduction of this paper we argue that the aim of a corporate insider disclosure regulation, as recognized by the authority, is to make private information available to all market participants more rapidly in order to promote price efficiency and market liquidity. In this Section we formally show that within the theoretical framework presented, the regulation fails at achieving these goals.

Our main result is that transparency reduces informational efficiency of prices and may cause the market to be less liquid. Before formalizing this in the next propositions, we first discuss the intuition behind. When the (risk averse) corporate insider is not required to disclose her trade, she will adopt a very aggressive trading strategy in order to exploit her informational advantage rapidly. The reason why a risk averse insider would behave more aggressively than the informed agent in Kyle (1985), is that the former wants to protect herself against future price risk imposed by liquidity traders. In other words, she is concerned about the possibility that profitable trading opportunities will be lost as the liquidity traders move market prices against her. By trading intensively at the early trading rounds, most of the private information is incorporated into prices, thus reducing the adverse selection problem faced by the market maker. Since the most of the private information is now in the market, the market maker reduces the sensitivity of prices with respect to the order

flows. This means a liquid market. Therefore, as stated by Holden and Subrahmanyam (1994), “from a regulatory perspective, [...] insider trading may be much less of a potential problem than the analysis of Kyle (1985) indicates”. It is the risk attitude of the informed agent that makes the market more efficient and liquid. In our benchmark model (the opaque market) the higher the risk aversion of the corporate insider, the faster private information get disseminated to the market, and the faster the market becomes liquid.

Surprisingly, the introduction of this transparency regulation prevents this effective “mechanism” based on risk attitude to succeed. Indeed, in equilibrium market prices convey less private information and liquidity does not decline over time. This represents a failure of the securities regulation. How can a transparency regulation worsen financial markets? Quoting Pagano and Röell (1996), this is possible “[...] because informed traders adapt their strategy to the market mechanism they face [...]”. In order to maintain an informational advantage over time, the corporate insider can not adopt the same trading strategy she would if no regulation was enforced, otherwise the market maker would be able to back out the private information from public records, thus eliminating any future profits. For this reason, we show in the model that an optimal strategy for the insider is to add a random component to her market order. On the one hand this allows her to maintain an informational advantage over time, but on the other hand it induces her to behave less aggressively. It is exactly this low trading intensity that reduces the dissemination of private information: trading prices have constant volatility over time, meaning that information is incorporated into prices at a constant rate. Moreover, since the insider must be indifferent to all possible values of her market order (otherwise she would deviate from that strategy), the adverse selection problem and hence market depth do not improve over time. Therefore, if a more realistic assumption on the risk preference of the informed agent increases informativeness of prices and reduces the adverse selection problem (Holden and Subrahmanyam, 1994; Baruch, 2002), the introduction of a trade disclosure regulation makes market prices less informative and the adverse selection problem more severe.

So, why do we observe mandatory trade disclosure regulation, if it can be so detrimental to financial markets? Our argument is that there are two opposite effects associated with the enforcement of such securities regulation – a positive *direct* effect and a negative *indirect* effect – and that regulators may have not considered or underestimated the latter. The positive direct effect is represented by the flow of information disclosed by the insider at the end of any trading round that clearly decreases the uncertainty caused by liquidity traders’ order flow. This by itself reduces the adverse selection problem of the market maker. Hence, a hasty conclusion may be: more information available to all market participants, less asymmetric information, more efficiency, more liquidity. However, the indirect and negative effect, due to the change in the insider’s trading strategy, intensifies the degree of asymmetric information. In this paper we show not only that an indirect effect also exists, but most importantly that it dominates the direct one.

Finally, another interesting interpretation of the effect of this insider trading regulation is that mandatory trade disclosure represents the institutional means that makes the informed agent behave as if she were risk neutral. To have a transparent market equilibrium in which the corporate insider maintains an informational advantage over time a non-invertible trading strategy is needed: adding a plain noise component to the market order of each auction is a tractable way to obtain one, as proposed by Huddart et al. (2001). However, to sustain this mixed strategy in equilibrium the trading costs must be constant over time, otherwise any disparity in such costs would induce a deviation from this strategy to exploit the lower cost. As a consequence, the insider trader's concern for future price risk is completely disregarded in equilibrium. The insider's risk aversion does not affect her trading behavior, thus making the transparent market equilibrium independent of its level. Therefore, the introduction of the disclosure regulation "forces" any risk averse insider to behave as a risk neutral one. Needless to say, the monopolistic risk neutral insider represents the most harmful case in terms of market efficiency and liquidity.⁹

As in Kyle (1985), the parameter $\Sigma(t)$, which gives the error variance of prices at time t , is an inverse measure of the informational efficiency of prices, which indicates how much of the insider's private information is not yet incorporated into prices. The parameter $\lambda(t)$, which characterizes the market maker pricing function, is an inverse measure of market liquidity (or more precisely of market depth). The parameter $\beta(t)$, which characterizes the insider's trading strategy, represents the intensity (aggressiveness) of insider trading, that is the sensitivity of the informed order flow to private information.

Panel A of Table 1 shows the effect of the disclosure regulation when the insider trader is assumed risk neutral. The three graphs contrast respectively $\Sigma(t)$, $\lambda(t)$, and $\beta(t)$, (i) in a transparent market (solid line), and (ii) in an opaque market (dashed line). This means plotting Huddart et al. (2001) versus Kyle (1985) in a continuous-time setting, in which exogenous parameters are normalized by setting $\Sigma(0) = \sigma_u^2 = 1$. In contrast to the results in a discrete-time setting (Huddart et al., 2001), market efficiency does not improve with the enforcement of the regulation. Insider trading intensity and conditional variance of prices coincides in the two market equilibria. The only difference relies on market liquidity, which is higher in a transparent market. The risk neutral case is very instructive because it clearly highlights the positive direct effect of the regulation aforementioned. Indeed, since insider aggressiveness remains unchanged, the negative indirect effect is null. This is what we believe being the reason why financial market regulators consider such regulation as beneficial for market participants.

However, Panel B of Table 1 shows that the results are reversed once we relax the restrictive assumption on the risk attitude. Specifically, we consider an informed agent characterized by a CARA utility function with coefficient of risk aversion of 4 and the same normalization on the

⁹In the case of a monopolistic risk neutral insider the trading costs are constant over time to eliminate profitable destabilization schemes that can generate unbounded profits, which would not be compatible with a rational expectation equilibrium.

exogenous parameters. The first graph shows that in an opaque market $\Sigma(t)$ declines very rapidly through time, whereas in a transparent market the instantaneous change in the conditional variance, $d\Sigma(t)$, is constant. Concerning the level, market efficiency is always higher in an opaque market. The third graph shows that in a transparent market insider trading aggressiveness is reduced at any auction. Finally, the second graph shows that with disclosure market depth is constant, while without disclosure it declines sharply over time, becoming lower in a final trading sub-period. In an opaque market the adverse selection problem is severe at the beginning of the trading period because the total order flow is very informative, and it becomes almost irrelevant once most of the information is incorporated into prices.

5.1 Market Efficiency

The normalized difference in conditional variance of prices between transparent and opaque market is a measure of the efficiency loss caused by the regulation, that is how much more information remains private and it is not incorporated into prices at each point in time if the disclosure regulation is introduced. An aggregate measure of efficiency loss is given by the sum of the efficiency losses at each point in time. Table 2 presents comparative statics for the instantaneous and the aggregate measure of efficiency loss with respect to the exogenous parameters of the model: the coefficient of risk aversion, A , the prior variance of the liquidation value of the risky asset, $\Sigma(0)$, and the variance of liquidity trading, σ_u^2 .

The three graphs in Panel A show that the instantaneous measure of efficiency loss is increasing in the level of risk aversion, and in both variances. For instance, according to the first graph, when the coefficient of risk aversion is equal to 4, the enforcement of the disclosure regulation prevents 57% of all private information to be incorporated into prices after just one-tenth of the trading rounds. When the coefficient of risk aversion is equal to 8, the instantaneous efficiency loss becomes 78%. Moreover, the higher the risk aversion the faster the private information reaches the market: this is confirmed by the right-skewness of the curves. Similar conclusions can be drawn with respect to the other two graphs.

Panel B, instead, considers the aggregate measure of efficiency loss and shows specular comparative statics. The aggregate efficiency loss is represented by the shaded area in the first graph (that is the sum of all the instantaneous losses) scaled by $1/2$ (that is the area below the solid line). Not surprisingly the second and the third graphs show that also the aggregate measure is increasing in all the three exogenous parameters. The following proposition formalizes these results.

Proposition 2. *For any level of risk aversion and at each trading round, insider trading is more*

aggressive and the market is more efficient in an opaque market:

$$\tilde{\beta}(t) - \beta(t) \geq 0 \quad \forall t \in [0, 1] \quad (29)$$

$$\tilde{\Sigma}(t) - \Sigma(t) \geq 0 \quad \forall t \in [0, 1] \quad (30)$$

If we define the “instantaneous efficiency loss” as the difference between the conditional variances of prices in a transparent and in an opaque market, normalized by the prior variance of the liquidation value of the risky asset, for every $t \in [0, 1]$,

$$\Omega(t) = \frac{\Sigma(t) - \tilde{\Sigma}(t)}{\Sigma(0)} \quad (31)$$

and the “aggregate efficiency loss” as the sum of all the instantaneous losses scaled by $1/2$,

$$\Omega^a = 2 \int_0^1 \Omega(t) dt \quad (32)$$

then the following cross-sectional results hold:

- i. $\Omega(t)$ increases with the level of the coefficient of risk aversion A ;
- ii. $\Omega(t)$ increases with the level of the prior variance of the liquidation value $\Sigma(0)$;
- iii. $\Omega(t)$ increases with the level of the instantaneous variance of liquidity trading σ_u^2 ;
- iv. the same comparative statics hold for Ω^a .

Proof. See the Appendix. □

5.2 Market Liquidity

In a similar fashion we define a measure of liquidity loss as the difference in the execution cost of liquidity traders between transparent and opaque market. Table 3 presents comparative statics with respect to the exogenous parameters of the model. All the three graphs in Panel A show that instantaneous liquidity loss is increasing over time, reflecting the change in the adverse selection problem. The aggregate liquidity loss is represented by the shaded area in the first graph of Panel B and it coincides with the difference in the aggregate execution cost, as defined by Back and Pedersen (1998). The second and the third graphs show that if the corporate insider is sufficiently risk averse, then the aggregate execution cost is higher in a transparent market. The following proposition formalizes the results on market liquidity.

Proposition 3. *If we define the “instantaneous liquidity loss” as the difference between the execution costs of liquidity traders in a transparent and in an opaque market for every $t \in [0, 1]$,*

$$\Psi(t) = (\lambda(t) - \tilde{\lambda}(t))\sigma_u^2 \quad (33)$$

and the “aggregate liquidity loss” as the sum of all the instantaneous losses,

$$\Psi^a = \int_0^1 \Psi(t)dt \quad (34)$$

then the following cross-sectional results hold:

- i. if the informed agent is sufficiently risk averse ($A > A^*$), then there exists trading sub-period in which market liquidity is higher in an opaque market, and the length of this sub-period increases with the level of the coefficient of risk aversion A ;*
- ii. if the informed agent is sufficiently risk averse ($A > A^{**} > A^*$), then Ψ^a increases with the level of the prior variance of the liquidation value, $\Sigma(0)$, and the level of the instantaneous variance of liquidity trading, σ_u^2 ;*
- iii. if the informed agent is sufficiently risk averse ($A > A^{***} > A^{**}$), then $\Psi^a > 0$.*

Proof. See the Appendix. □

6 Insider Trading Profitability

To this point we have analyzed the implications of insider trade disclosure regulation for market efficiency and liquidity, since identified by the regulator as being the two main objective of such regulation. In the current section we describe some findings concerning the insider ex-ante expected profits.¹⁰

In order to have a more flexible specification for the utility function, which encompasses the risk neutral case when the coefficient of risk aversion goes to zero, let us consider the following functional form,

$$U(W(1)) = \frac{1 - \exp\{-AW(1)\}}{A} \quad (35)$$

Note that the equilibrium results are the same regardless of the use of the utility functions specified in Equations (2) and (35) because we can take a monotonic transformation of utility and still represent the same preferences.¹¹ Now, let π_0 denote the insider’s unconditional expected utility

¹⁰I thank Eric Hughson for inspiring the development of this section.

¹¹With the utility function specified in Equation (35), $\lim_{A \rightarrow 0} U(W(1)) = W(1)$.

(at time 0),

$$\pi_0 = \mathbb{E}_0[U(W(1))] \quad (36)$$

and Π_0 the insider's expected utility conditional on her private information (still at time 0, that is before the trading takes place),

$$\Pi_0 = \mathbb{E}_0[U(W(1))|v] \quad (37)$$

where $\pi_0 = \mathbb{E}_0[\Pi_0]$ by the Law of Iterated Expectation.

The following proposition examines how the insider's (un)conditional expected utility changes across the two regulatory regimes, the *opaque* and the *transparent* market, focusing in particular on the role played by her risk attitude.

Proposition 4. *In an opaque market equilibrium the insider's unconditional and conditional expected utility are given by*

$$\tilde{\pi}_0 = \frac{1}{A} \left[1 - \left(1 + A \left(A\Sigma(0)\sigma_u^2 + \sqrt{\Sigma(0)\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)} \right) \right)^{-1/2} \exp \left\{ -AW(0) \right\} \right] \quad (38)$$

$$\tilde{\Pi}_0 = \frac{1}{A} \left[1 - \Xi \exp \left\{ -A \left[W(0) + \frac{1}{4} \left(\sqrt{\frac{\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)}{\Sigma(0)}} - A\sigma_u^2 \right) (v - \mathbb{E}_0(v))^2 \right] \right\} \right] \quad (39)$$

respectively, where

$$\Xi \equiv \left(\frac{2}{2 + A \left(A\Sigma(0)\sigma_u^2 + \sqrt{\Sigma(0)\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)} \right)} \right)^{1/2} \quad (40)$$

while in a transparent market equilibrium they are given by

$$\pi_0 = \frac{1}{A} \left[1 - \left(1 + A\sqrt{\Sigma(0)\sigma_u^2} \right)^{-1/2} \exp \left\{ -AW(0) \right\} \right] \quad (41)$$

$$\Pi_0 = \frac{1}{A} \left[1 - \exp \left\{ -A \left[W(0) + \left(\frac{1}{2} \sqrt{\frac{\sigma_u^2}{\Sigma(0)}} \right) (v - \mathbb{E}_0(v))^2 \right] \right\} \right] \quad (42)$$

Then the following results hold:

i. $\tilde{\pi}_0 > \pi_0$ and $\partial(\tilde{\pi}_0 - \pi_0)/\partial A \leq 0$;

ii. if we define the insider's initial informational advantage as $\Gamma \equiv |v - \mathbb{E}_0(v)|$, then the set

$$\mathcal{G} = \{\Gamma : \Pi_0(\Gamma) \geq \tilde{\Pi}_0(\Gamma)\}$$

is non-empty if the insider is risk averse ($A > 0$), and takes the form of open intervals $[\Gamma^*, \infty)$

on $\mathbf{R}^+$; Γ^* is increasing in $\Sigma(0)$, decreasing in σ_u^2 , and decreasing in A ; in the limit, as A approaches 0, Γ^* goes to infinity.

Proof. See the Appendix. □

The main finding can be summarized as follows: in a transparent market, compared to an opaque market, the insider's *unconditional* expected utility is reduced; however, if she is risk averse, once the private information is realized, her *conditional* expected utility may be higher. This is the case when the unexpected component of the private information is sufficiently large.

The first part of the proposition states that the introduction of a transparency regulation reduces the ex-ante expected utility of the corporate insider. However, the difference in utility across the two regulatory regimes becomes smaller and smaller the more averse to the risk the insider is. We describe these *unconditional* results in Panel A of Table 4. The three graphs contrast the unconditional expected utility of the insider over the prior variance of the private information (the first two graphs) and over the level of risk aversion (third graph). The dashed line, which represents an opaque market, is always above the solid line, a transparent market, while the solid bold line, which captures the difference between the two is clearly decreasing in A .

The second part of the proposition highlights a more interesting feature of the model, that further differentiates the general case of risk aversion from the specific case of risk neutrality. With risk neutrality no matter what type of information the insider receives, she is worse off if she has to disclose her trade; with risk aversion, instead, if the private information is particularly unexpected, the insider can indeed be better off. This result is important because it shows that there are states of the world in which the insider can benefit from the introduction of such regulation. Therefore, trade disclosure not only fails at improving market efficiency and liquidity, but it may also allow the informed agent to exploit her private information in a more profitable way. To understand why this is possible, let us consider the value function at time 0, given zero initial wealth:

$$\Pi_0 = \frac{1}{A} \left[1 - \exp \left\{ -A \left(\alpha(0) [M(0)]^2 + \delta(0) \right) \right\} \right] \quad (43)$$

where $M(0) = v - p(0)$ represents the component of the private information which is not expected by the market.¹²

The coefficient $\alpha(0)$ captures the initial liquidity of the market, which in equilibrium depends on the aggressiveness of the insider. In a transparent market the insider adopts a mixed strategy which makes her trade less aggressive, thus inducing a more liquid market. In an opaque market, instead, the market maker anticipates the willingness of the insider to exploit the private information in a timely manner, and hence sets a price which is very sensitive to the order flow: this makes the

¹²Note that the insider's initial informational advantage $\Gamma = |M(0)|$.

market illiquid. Indeed, straightforward algebra shows that $\alpha(0) \geq \tilde{\alpha}(0)$, where the equality holds only for the case of risk neutrality.

The coefficient $\delta(0)$ reflects the possibility for the insider to profit by hiding her trades behind the ones placed by liquidity traders for the entire trading period. It is the “aggregate value” of noise trading activities for the insider. This is positive in an opaque market since the insider’s future informational advantages $|v - p(t)|$ depend on $du(t)$. In particular $\tilde{\delta}(0)$ is proportional to the “quantity” of noise trading activities, measured by the variance σ_u^2 , times the “price” of such activities, measured by the coefficients $\tilde{\lambda}(t)$, for all t , which capture impact of noise trades on market prices. Not surprisingly, in a transparent market the coefficient $\delta(0)$ is equal to zero. As a matter of fact the insider’s future informational advantages $|v - p^*(t)|$ do not depend on $du(t)$ because the disclosure of her trades eliminates the noise trading uncertainty faced by the market maker. It follows that $\delta(0) < \tilde{\delta}(0)$.

Qualitatively, we can interpret $[\tilde{\delta}(0) - \delta(0)]$ as a measure of the positive *direct* effect, and $[\tilde{\alpha}(0) - \alpha(0)]$ as a measure as the negative *indirect* effect brought by the introduction of the trade disclosure regulation. With risk neutrality, $\tilde{\alpha}(0) = \alpha(0)$, and only the positive effect remains.

Having examined the role played by the two components that characterize the value function at time 0, now it should be clear that the relative weight of these components determines whether the insider’s conditional expected utility is higher in a transparent rather than in an opaque market. As appears in Equation (43), the relative weight is given by the (square of the) insider’s informational advantage. The lower Γ , the lower the importance of the negative effect, and the lower the insider’s expected utility. This justifies the existence of a threshold level Γ^* , such that the insider’s expected utility is higher in a transparent market for any level of the informational advantage greater than Γ^* . By continuity, when the informational advantage equal to Γ^* , the expected utilities associated to the two regulatory regimes coincide. Panel B in Table 4 highlights these results. The first two graphs describe how the conditional expected utility changes with Γ for the case of risk neutrality and risk aversion respectively. When A approaches zero, $\tilde{\alpha}(0)$ approaches $\alpha(0)$, and $\tilde{\delta}(0)$ approaches its maximum: therefore, the threshold level Γ^* must be very high. In the limit, Γ^* goes to infinity, and this is the reason why it does not appear in the first graph. With risk aversion, Γ^* is finite, as pictured in the second graph. Finally, the third graph presents comparative statics of Γ^* for the exogenous parameters A , $\Sigma(0)$, and σ_u^2 .

The current section provides interesting findings concerning the profitability of the informed trading activity for corporate insiders. A more rigorous welfare analysis should also take into account the impact of the disclosure regulation on the welfare of liquidity traders. However, in the framework here presented, the preferences of agents who trade for liquidity purposes are unmodeled and their aggregate trade is exogenously assumed. Departures from this assumption are analyzed in Admati and Pfleiderer (1988), Bernhardt and Hughson (1997), and Mendelson and Tunca (2004) among others.

7 Conclusions and Policy Implications

Securities markets worldwide have different degrees of transparency with implications that are not well understood. In this article we focus on mandatory ex-post trade disclosure by corporate insiders as an important aspect of market transparency. In a continuous-time model of risk averse strategic insider trading we show that informational efficiency and market liquidity are significantly lower in a transparent market (i.e. with disclosure regulation) than in an opaque market (i.e. without disclosure regulation). The reason for this detrimental effect is that a risk averse insider optimally chooses a less aggressive trading strategy in order to prevent the market maker from inferring perfectly the private information from public records, and to maintain her informational advantage over time. Moreover, if the initial informational advantage of the insider is sufficiently high, then her expected utility from trading is higher in a transparent market.

Our result adds an interesting theoretical evidence to the existing debate on the relation between the degree of transparency and the optimal design and regulation of securities markets. A number of implications for regulatory policy can be drawn from our analysis. First, a mandatory insider trade disclosure does not eliminate the presence of insider trading once the informed agent reveals her trade. Second, if the main goal of market design is to sustain informational efficiency and liquidity, as explicitly stated by several financial regulations such as the Section 16(a) of the U.S. Securities Exchange Act, then an opaque market should be preferred. Strategic corporate insiders would exploit their informational advantage more rapidly, inducing private information to reach the market (and hence to be available to all investors) on a more timely basis. Under this view a trade disclosure regulation would represent a friction in the system of efficient prices. Such regulation may also not be able to sufficiently reduce the adverse selection problem in the market, and consequently to enhance market liquidity. Finally, when the degree of asymmetric information is high – that is when an effective policy intervention is more needed – the introduction of a trade disclosure regulation increases the insider trading profitability. Testable implications on insiders' trading behavior and on market characteristics then follow.

The change in the U.S. regulation on insider trade disclosure with the introduction of the Sarbanes-Oxley Act in 2002 offers the opportunity to carry out a natural experiment to test the implications derived in this article. Moreover, a more general formulation of our model, as highlighted in Remark 1, would allow us to study optimal disclosure delays in a dynamic setting. These interesting and challenging projects are left for future research.

Appendix A: Proofs

Proof of Theorem 1. The objective of the corporate insider is to maximize

$$\max_{dx(t)} \mathbb{E}[-\exp\{-AW(1)\}] \quad (\text{A.1})$$

where the wealth of the insider at time t is given by

$$W(t) = \int_0^t (v - (p(s) + dp(s))) dx(s) \quad (\text{A.2})$$

as discussed in Back (1992). The insider's optimization problem entails solving a Markovian stochastic control problem, in which the HJB equation is given by

$$0 = \max_{dx} \mathbb{E}_t[dJ(M, t)] \quad (\text{A.3})$$

Let $K(M, t) \equiv W(t) + V(M, t)$. Given the conjectured value function in Equation (8), it follows that:¹³

$$dJ(M, t) = -AJ(M, t) \left\{ dK(M, t) - \frac{A}{2} d\langle K, K \rangle_t \right\} \quad (\text{A.4})$$

and

$$\mathbb{E}_t[dJ(M, t)] = -AJ(M, t) \left\{ \mathbb{E}_t[dK(M, t)] - \frac{A}{2} \text{Var}_t[dK(M, t)] \right\} \quad (\text{A.5})$$

since $(\mathbb{E}_t[dK(M, t)])^2 = 0$. Therefore, the HJB equation can be written as

$$0 = \max_{dx} \mathbb{E}_t[dK(M, t)] - \frac{A}{2} \text{Var}_t[dK(M, t)] \quad (\text{A.6})$$

Let $M(t) \equiv v - p(t)$, by Ito's lemma:¹⁴

$$\begin{aligned} dW(t) &= M(t)dx(t) - \lambda(t)d\langle x, x \rangle_t - \lambda(t)d\langle u, x \rangle_t \\ &= M(t)dx(t) \end{aligned} \quad (\text{A.7})$$

$$dM(t) = -\tilde{\lambda}(t)[dx(t) + du(t)] \quad (\text{A.8})$$

$$d\langle M, M \rangle_t = \tilde{\lambda}(t)^2 \sigma_u^2 dt \quad (\text{A.9})$$

$$\begin{aligned} dV(M, t) &= V_M dM(t) + \frac{1}{2} V_{MM} d\langle M, M \rangle_t + V_t dt \\ &= [2\tilde{\alpha}(t)M(t)]dM(t) + [\tilde{\alpha}(t)]d\langle M, M \rangle_t + M(t)^2 d\tilde{\alpha}(t) + d\tilde{\delta}(t) \end{aligned} \quad (\text{A.10})$$

¹³The differential $d\langle X, Y \rangle_t$ coincides with $dX(t)dY(t)$ and represents the differential of the quadratic variation process.

¹⁴Note that: (i) $d\langle u, x \rangle_t = 0$ since $d\langle B^u, B^x \rangle_t = 0$ (the two Brownian motion are assumed independent) and $dB^u dt$ is of order higher than dt ; (ii) $d\langle x, x \rangle_t = 0$ because of order $(dt)^2$.

The differential $dK(M, t) = dW(t) + dV(M, t)$ is random because of $du(t)$, and hence normally distributed. Given the conjectured trading strategy in Equation (5), Equation (A.6) can be written as

$$0 = \max_{\tilde{\beta}(t)} \tilde{\beta}(t) \left[1 - 2\tilde{\alpha}(t)\tilde{\lambda}(t) \right] M(t)^2 dt + \tilde{\alpha}(t)\tilde{\lambda}(t)^2 \sigma_u^2 dt + d\tilde{\delta}(t) + \left[d\tilde{\alpha}(t) - 2A\tilde{\alpha}(t)^2 \tilde{\lambda}(t)^2 \sigma_u^2 dt \right] M(t)^2 \quad (\text{A.11})$$

The FOC with respect to $\tilde{\beta}(t)$ is given by

$$\left[1 - 2\tilde{\alpha}(t)\tilde{\lambda}(t) \right] M(t)^2 dt = 0 \quad (\text{A.12})$$

and it is satisfied for any $t \in [0, 1]$ and any $M(t)$ if and only if

$$\tilde{\alpha}(t)\tilde{\lambda}(t) = \frac{1}{2} \quad (\text{A.13})$$

Substituting Equation (A.13) into Equation (A.11) we obtain

$$0 = \frac{1}{2} \tilde{\lambda}(t) \sigma_u^2 dt + d\tilde{\delta}(t) + \left[d\tilde{\alpha}(t) - \frac{A}{2} \sigma_u^2 dt \right] M(t)^2 \quad (\text{A.14})$$

which holds for any $M(t)$ if and only if:

$$d\tilde{\alpha}(t) = \frac{A}{2} \sigma_u^2 dt \quad (\text{A.15})$$

$$d\tilde{\delta}(t) = -\frac{1}{2} \tilde{\lambda}(t) \sigma_u^2 dt \quad (\text{A.16})$$

Given the equilibrium condition in Equation (A.13) and the dynamics of $\tilde{\alpha}(t)$ in Equation (A.15), it follows that

$$d\tilde{\lambda}(t) = -A\tilde{\lambda}(t)^2 \sigma_u^2 dt \quad (\text{A.17})$$

with solution equal to

$$\tilde{\lambda}(t) = (A\sigma_u^2 t + k)^{-1} \quad (\text{A.18})$$

for some constant k . Moreover, since no utility can be gained after trading is complete, the boundary condition $\tilde{\delta}(1) = 0$ must hold. Therefore,

$$\tilde{\delta}(t) = \frac{\sigma_u^2}{2} \int_t^1 \tilde{\lambda}(s) ds \quad (\text{A.19})$$

Following closely Back et al. (2000), we now define in the following Lemma the filtering problem of the market maker, whose information structure is represented by the filtration $\mathbb{F} \equiv \{\mathcal{F}(t) | 0 \leq t < 1\}$ generated by the total order flow $y(t)$.

Lemma 1. *Assume the insider trader follow the linear strategy as in Equation (5) and let us define $P(t) \equiv$*

$\mathbb{E}[v|\mathcal{F}(t)]$ and $\Sigma(t) \equiv \text{Var}[v|\mathcal{F}(t)]$. Then the following process

$$Q(t) = \int_0^t \tilde{\beta}(s)[v - P(s)]ds + u(t) \quad (\text{A.20})$$

is a Wiener process on the market maker's information structure $\mathbb{F}$ and it is called "innovation" process for the market maker's estimation problem. The differential

$$dQ(t) = \tilde{\beta}(t)[v - P(t)]dt + du(t) \quad (\text{A.21})$$

is the unpredictable part of the total order flow. Moreover,

$$P(t) = \int_0^t \frac{\tilde{\beta}(s)\tilde{\Sigma}(s)}{\sigma_u^2} dQ(s) \quad (\text{A.22})$$

The market maker's estimate of v is revised according to

$$dP(t) = \frac{\tilde{\beta}(t)\tilde{\Sigma}(t)}{\sigma_u^2} dQ(t) \quad (\text{A.23})$$

Finally,

$$d\tilde{\Sigma}(t) = -\frac{[\tilde{\beta}(t)\tilde{\Sigma}(t)]^2}{\sigma_u^2} dt \quad (\text{A.24})$$

Proof. This is an application of the Kalman-Bucy filter. See Kallianpur (1980, Sec. 10.3). $\square$

Comparing Equation (A.23) with Equation (6) we can conclude that

$$\tilde{\lambda}(t) = \frac{\tilde{\beta}(t)\tilde{\Sigma}(t)}{\sigma_u^2} \quad (\text{A.25})$$

Given our linear choice space, $\lim_{s \rightarrow 1} p(s) = v$, meaning that all the private information is incorporated into prices at the end of the trading game. Therefore, $\Sigma(1) = 0$. Considering the conditional variance process specified in Equation (A.24), it follows that

$$\tilde{\Sigma}(t) = \int_t^1 \tilde{\lambda}(s)^2 \sigma_u^2 ds \quad (\text{A.26})$$

for every $t \in [0, 1]$. A closed-form solution for this equilibrium is obtained by substituting Equation (A.18) in the expression for $\tilde{\Sigma}(t)$ evaluated at $t = 0$,

$$\Sigma(0) = \int_0^1 \tilde{\lambda}(s)^2 \sigma_u^2 ds \quad (\text{A.27})$$

$$= \int_0^1 (A\sigma_u^2 s + k)^{-2} \sigma_u^2 ds \quad (\text{A.28})$$

$$= \frac{\sigma_u^2}{k(k + A\sigma_u^2)} \quad (\text{A.29})$$

and solving for (the positive root of) k :

$$k = \frac{\sqrt{\Sigma(0)\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)} - A\Sigma(0)\sigma_u^2}{2\Sigma(0)} \quad (\text{A.30})$$

Substituting this last expression back into Equation (A.18), we obtain Equation (11).¹⁵ Similarly, evaluating the integral in Equation (A.26) and considering that $\tilde{\beta}(t) = \tilde{\lambda}(t)\sigma_u^2/\tilde{\Sigma}(t)$, we get Equation (10) and (12) respectively. This completes the proof of the theorem. $\square$

Proof of Theorem 2. We follow closely the Proof of Theorem 1. Given the conjectured value function defined in Equation (20), and considering the same steps from Equation (A.3) to Equation (A.6), the HJB equation is given by

$$0 = \max_{dx} \mathbb{E}_t[dK(M, t)] - \frac{A}{2} \text{Var}_t[dK(M, t)] \quad (\text{A.32})$$

where $K(M, t) \equiv W(t) + V(M, t)$. Let $M(t) \equiv v - p^*(t)$, by Ito's lemma:¹⁶

$$\begin{aligned} dW(t) &= M(t)dx(t) - \lambda(t)d\langle x, x \rangle_t - \lambda(t)d\langle u, x \rangle_t \\ &= M(t)dx(t) - \lambda(t)d\langle x, x \rangle_t \end{aligned} \quad (\text{A.33})$$

$$dM(t) = -\gamma(t)dx(t) \quad (\text{A.34})$$

$$d\langle M, M \rangle_t = \gamma(t)^2 d\langle x, x \rangle_t \quad (\text{A.35})$$

$$\begin{aligned} dV(M, t) &= V_M dM(t) + \frac{1}{2} V_{MM} d\langle M, M \rangle_t + V_t dt \\ &= [2\alpha(t)M(t)]dM(t) + [\alpha(t)]d\langle M, M \rangle_t + M(t)^2 d\alpha(t) + d\delta(t) \end{aligned} \quad (\text{A.36})$$

The differential $dK(M, t) = dW(t) + dV(M, t)$ is random because of dissimulation component of the informed order flow $dz(t)$, and hence normally distributed. Given the conjectured trading strategy in Equation (15), Equation (A.32) can be written as

$$\begin{aligned} 0 = \max_{\beta(t)} & \beta(t) \left[1 - 2\alpha(t)\gamma(t) \right] M(t)^2 dt + \sigma_z^2(t) \left[(\alpha(t)\gamma(t)^2 - \lambda(t)) - \frac{A}{2} (1 - 2\alpha(t)\gamma(t))^2 M(t)^2 \right] dt \\ & + M(t)^2 d\alpha(t) + d\delta(t) \end{aligned} \quad (\text{A.37})$$

The FOCs with respect to $\beta(t)$ and $\sigma_z^2(t)$ are given by

$$(1 - 2\alpha(t)\gamma(t)) M(t)^2 = 0 \quad (\text{A.38})$$

$$(\alpha(t)\gamma(t)^2 - \lambda(t)) - \frac{A}{2} (1 - 2\alpha(t)\gamma(t))^2 M(t)^2 = 0 \quad (\text{A.39})$$

¹⁵An alternative way to determine the equilibrium $\tilde{\lambda}(t)$ is to express it as a function of $\tilde{\lambda}(1)$ (instead of k), and then solve the following system of non linear equations for the positive roots $\{\tilde{\lambda}(t), \tilde{\lambda}(1)\}$:

$$\begin{cases} \tilde{\lambda}(t) &= \frac{\tilde{\lambda}(1)}{1 - A\sigma_u^2 \tilde{\lambda}(1)(1-t)} \\ \Sigma(0) &= \int_0^1 \left(\frac{\tilde{\lambda}(1)\sigma_u}{1 - A\sigma_u^2 \tilde{\lambda}(1)(1-s)} \right)^2 ds \end{cases} \quad (\text{A.31})$$

¹⁶Note that: (i) $d\langle u, x \rangle_t = 0$ since $d\langle B^u, B^z \rangle_t = 0$ and $dB^u dt$ is of order higher than dt ; (ii) $d\langle x, x \rangle_t \neq 0$ because of the dissimulation component $dz(t)$.

respectively. These conditions are satisfied for any $M(t)$ if and only if

$$1 - 2\alpha(t)\gamma(t) = 0 \quad (\text{A.40})$$

$$\alpha(t)\gamma(t)^2 - \lambda(t) = 0 \quad (\text{A.41})$$

Note that these conditions guarantee that in equilibrium the insider is indifferent across all possible values of her trade (requirement for an equilibrium in mixed-strategies). Substituting Equations (A.40) and (A.41) into Equation (A.37) we obtain

$$0 = M(t)^2 d\alpha(t) + d\delta(t) \quad (\text{A.42})$$

which holds for any $M(t)$ if and only if $\alpha(t)$ and $\delta(t)$ are constant:

$$d\alpha(t) = 0 \quad (\text{A.43})$$

$$d\delta(t) = 0 \quad (\text{A.44})$$

Given the boundary condition $\tilde{\delta}(1) = 0$ (no utility can be gained after trading is complete), $\delta(t) = 0$ for all $t \in [0, 1]$. Moreover, since $\alpha(t)$ is constant, by Equation (A.40) it must be the case that also $\gamma(t)$ is constant. The same holds true for $\lambda(t)$ once considered Equation (A.41). Therefore,

$$\hat{\lambda} = \frac{1}{2}\hat{\gamma} \quad (\text{A.45})$$

$$\hat{\alpha} = \frac{1}{2}\hat{\gamma}^{-1} \quad (\text{A.46})$$

where $\{\hat{\gamma}, \hat{\lambda}, \hat{\alpha}\}$ denote the constant equilibrium values for $\{\gamma(t), \lambda(t), \alpha(t)\}$, respectively, for all $t \in [0, 1]$.

A simple application of Lemma 1 allows us to characterize the filtering problems faced by the market maker, by considering both filtration $\mathbb{F} \equiv \{\mathcal{F}(t) | 0 \leq t < 1\}$ generated by the total order flow $y(t)$ and past disclosures (*pre-disclosure filtration*) and the filtration $\mathbb{F}' \equiv \{\mathcal{F}'(t) | 0 \leq t < 1\}$ generated by the insider's order flow $x(t)$ (*post-disclosure filtration*). Given the market efficiency conditions $p(t) = \mathbb{E}_t[v|\mathcal{F}(t)]$, $p_t^* = \mathbb{E}_t[v|\mathcal{F}'(t)]$ and the conjectured pricing rules in Equations (17) and (18), it follows that

$$\gamma(t) = \frac{\beta(t)\Sigma(t)}{\sigma_z^2(t)} \quad (\text{A.47})$$

$$\lambda(t) = \frac{\beta(t)\Sigma(t)}{\sigma_z^2(t) + \sigma_u^2} \quad (\text{A.48})$$

Moreover, given $\Sigma(t) = \mathbb{V}\text{ar}[v|\mathcal{F}'(t)]$, we obtain

$$d\Sigma(t) = -\frac{[\beta(t)\Sigma(t)]^2}{\sigma_z^2(t)} dt \quad (\text{A.49})$$

Substituting Equations (A.47) and (A.48) into Equations (A.40) and (A.41), we can solve for the optimal dissimulation component as follows:

$$\lambda(t) = \frac{1}{2}\gamma(t) \Rightarrow \frac{\beta(t)\Sigma(t)}{\sigma_z^2(t) + \sigma_u^2} = \frac{1}{2} \frac{\beta(t)\Sigma(t)}{\sigma_z^2(t)} \Rightarrow \sigma_z^2(t) = \sigma_u^2 \quad (\text{A.50})$$

Given the constant drift for the conditional variance process, it follows that

$$\Sigma(t) = \Sigma(0) - \int_0^t \gamma(s)^2 \sigma_z(s)^2 ds \quad (\text{A.51})$$

$$= \Sigma(0) - (\hat{\gamma}^2 \sigma_u^2) t \quad (\text{A.52})$$

for every $t \in [0, 1]$, and given our linear choice space ($\lim_{s \rightarrow 1} p^*(s) = \lim_{s \rightarrow 1} p(s) = v$), all the private information is incorporated into prices at the end of the trading game: $\Sigma(1) = 0$. Then, it follows that

$$\hat{\gamma} = (\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (\text{A.53})$$

All the other constants ($\beta(t)$, $\lambda(t)$, $\alpha(t)$) are obtained through straightforward algebra. The boundary condition $\sigma_z(1) = 0$ states that no dissimulation component is added to the trading strategy at the last trading instant. This completes the proof of the theorem. $\square$

Proof of Proposition 1. Once the positive root of the polynomial in Equation (B.11) is determined, the sequential auction equilibrium is fully characterized for any number of auctions N . It is straightforward to see that for $\Delta t \rightarrow 0$ the polynomial becomes of order two and the only positive root λ is equal to $\sqrt{\Sigma(0)/4\sigma_u^2}$. Since $t \in [0, 1]$, then $n = tN$ and the following limits (for $\Delta t \rightarrow 0$) hold true:

$$\gamma_n \rightarrow (\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (\text{A.54})$$

$$\Lambda \rightarrow (\Sigma(0)/\sigma_u^2)^{\frac{1}{2}} \quad (\text{A.55})$$

$$\beta_n \rightarrow \sigma_u \Sigma(0)^{-\frac{1}{2}} / (1 - t) \quad (\text{A.56})$$

$$\sigma_{z_n}^2 \rightarrow \sigma_u^2 \quad (\text{A.57})$$

$$\alpha_n \rightarrow \frac{1}{2} (\sigma_u^2 / \Sigma(0))^{\frac{1}{2}} \quad (\text{A.58})$$

$$\Sigma_n \rightarrow (1 - t) \Sigma(0) \quad (\text{A.59})$$

$$\delta_n \equiv \ln(\eta_n) \rightarrow 0 \quad (\text{A.60})$$

$\square$

Proof of Proposition 2. $\Sigma(t) - \tilde{\Sigma}(t) \geq 0$ and $\tilde{\beta}(t) - \beta(t) \geq 0$ imply respectively

$$\frac{A[\Sigma(0)]^{\frac{3}{2}} \sigma_u \left[\sqrt{4 + A^2 \Sigma(0) \sigma_u^2} + \sqrt{A^2 \Sigma(0) \sigma_u^2 (1 - 2t)} \right] (1 - t)t}{[2 + 2A^2 \Sigma(0) \sigma_u^2 (1 - t)t]} \geq 0 \quad (\text{A.61})$$

$$\frac{A \Sigma(0) \sigma_u^2 + \sqrt{\Sigma(0)} \sigma_u \left[\sqrt{4 + A^2 \Sigma(0) \sigma_u^2} - 2 \right]}{2 \Sigma(0) (1 - t)} \geq 0 \quad (\text{A.62})$$

Straightforward algebra can show that these two inequalities hold true for any $t \in [0, 1]$. Using the first result we can derive $\Omega(t)$ simply by dividing the l.h.s of Equation (A.61). The partial derivative of $\Omega(t)$ with

respect to the coefficient of risk aversion is equal to

$$\frac{\partial \Omega(t)}{\partial A} = \frac{\sqrt{\Sigma(0)\sigma_u^2} \left[2 + A\sqrt{\Sigma(0)\sigma_u^2} \sqrt{4 + A^2\Sigma(0)\sigma_u^2} (1-t) + A^2\Sigma(0)\sigma_u^2 (1+2(t-1)t) \right] (1-t)t}{\sqrt{4 + A^2\Sigma(0)\sigma_u^2} \left[1 + A^2\Sigma(0)\sigma_u^2 (1-t)t \right]^2} \quad (\text{A.63})$$

Straightforward algebra can show that this partial derivative is non-negative for any $t \in [0, 1]$. This proves part (i). Part (ii) and (iii) follow since

$$\frac{\partial \Omega(t)}{\partial \Sigma(0)} = \frac{\partial \Omega(t)}{\partial A} \frac{A}{2} \frac{1}{\Sigma(0)} \quad (\text{A.64})$$

$$\frac{\partial \Omega(t)}{\partial \sigma_u^2} = \frac{\partial \Omega(t)}{\partial A} \frac{A}{2} \frac{1}{\sigma_u^2} \quad (\text{A.65})$$

Finally, a simple application of Leibniz's rule proves part (iv). $\square$

Proof of Proposition 3. Since $\tilde{\lambda}(t)$ is decreasing over time,

$$\frac{\partial \tilde{\lambda}(t)}{\partial t} = - \frac{4A\Sigma(0)}{\left(\sqrt{4 + A^2\Sigma(0)\sigma_u^2} + A\sigma_u\sqrt{\Sigma(0)}(2t-1) \right)^2} \quad (\text{A.66})$$

this means that $\tilde{\lambda}(t)$ and $\lambda(t)$ cross at most once in $t \in [0, 1]$. Moreover, since $\tilde{\lambda}(1)$ is decreasing in the level of risk aversion,

$$\left. \frac{\partial \tilde{\lambda}(t)}{\partial A} \right|_{t=1} = - \frac{\Sigma(0)}{2} \left(1 - \sqrt{\frac{A^2\Sigma(0)\sigma_u^2}{4 + A^2\Sigma(0)\sigma_u^2}} \right) \quad (\text{A.67})$$

then for any $A > A^*$, with $A^* = 3 \left[2\sqrt{\Sigma(0)\sigma_u^2} \right]^{-1}$, market liquidity is higher in an opaque market for the trading sub-period $[t^*, 1]$, with

$$t^* = \frac{A\Sigma(0)\sigma_u + \sqrt{\Sigma(0)}(4 - \sqrt{4 + A^2\Sigma(0)\sigma_u^2})}{2A\Sigma(0)\sigma_u} \quad (\text{A.68})$$

Moreover $\partial t^*/\partial A < 0$: this proves part (i). The aggregate liquidity loss is equal to

$$\Psi^a = \frac{\sqrt{\Sigma(0)\sigma_u^2}}{2} + \frac{1}{A} \ln \left(\frac{\sqrt{4 + A^2\Sigma(0)\sigma_u^2} - A\sqrt{\Sigma(0)\sigma_u^2}}{\sqrt{4 + A^2\Sigma(0)\sigma_u^2} + A\sqrt{\Sigma(0)\sigma_u^2}} \right) \quad (\text{A.69})$$

Straightforward algebra can show that for any $A > A^{**} > A^*$, with $A^{**} = 2\sqrt{3} \left[\sqrt{\Sigma(0)\sigma_u^2} \right]^{-1}$, the partial derivatives $\partial \Psi^a / \partial \Sigma(0)$ and $\partial \Psi^a / \partial \sigma_u^2$ are both positive. This proves part (ii). Finally, since Ψ^a is monotone increasing in A ($\partial \Psi^a / \partial A > 0$ for any level of A) and since $\Psi^a(A^{**}) < 0$ and $\lim_{A \rightarrow \infty} \Psi^a = \sqrt{\Sigma(0)\sigma_u^2}/2$, then there must exist a level $A^{***} > A^{**}$ such that $\Psi^a > 0$ for any $A > A^{***}$. This concludes the proof of the proposition. $\square$

Proof of Proposition 4. Given the monotonic transformation, $T(x) = (1+x)/A$, of the utility function, as in Equation (35), the insider's conditional utility at time zero (conditional on the private information v)

is given by

$$\Pi_0 = \frac{1}{A} [1 + J(M, 0)] = \frac{1}{A} [1 - \exp\{-A[W(0) + \alpha(0)M(0)^2 + \delta(0)]\}] \quad (\text{A.70})$$

Substituting the equilibrium values of $\alpha(0)$ and $\delta(0)$ (Equations (26) and (27) respectively) into Equation (A.70), we obtain Equation (42). Similarly, if we substitute $\tilde{\alpha}(0)$ and $\tilde{\delta}(0)$ (Equations (13) and (14) respectively) into Equation (A.70), we obtain the expression in Equation (39). Taking expectation with respect to the random variable $M(0) \sim \mathcal{N}(0, \Sigma(0))$ of Equations (39) and (42), we obtain Equations (38) and (41) respectively.¹⁷

$$\tilde{\pi}_0 = \frac{1}{A} [1 - \tilde{\Upsilon} \exp\{-AW(0)\}] \quad (\text{A.71})$$

$$\pi_0 = \frac{1}{A} [1 - \Upsilon \exp\{-AW(0)\}] \quad (\text{A.72})$$

where

$$\tilde{\Upsilon} = \left(1 + A \left(A\Sigma(0)\sigma_u^2 + \sqrt{\Sigma(0)\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)}\right)\right)^{-1/2} \quad (\text{A.73})$$

$$\Upsilon = \left(1 + A\sqrt{\Sigma(0)\sigma_u^2}\right)^{-1/2} \quad (\text{A.74})$$

It is easy to see that for any finite and positive A , $\tilde{\Upsilon} < \Upsilon$, and hence that $\tilde{\pi}_0 > \pi_0$. Moreover, we can conclude that $\partial(\tilde{\pi}_0 - \pi_0)/\partial A \leq 0$ by the following factors: (i) $\nexists$ a finite and positive value of A such that $\partial(\tilde{\pi}_0 - \pi_0)/\partial A = 0$; and (ii) $\lim_{A \rightarrow 0} \partial(\tilde{\pi}_0 - \pi_0)/\partial A = -(5\Sigma(0)\sigma_u^2 + 4\sqrt{\Sigma(0)\sigma_u^2}W_0)/8$. Straightforward algebra confirms the result.

Solving the equation $(\tilde{\Pi}_0 - \Pi_0) = 0$ for $M(0)$, we find:

$$M^*(0) = \pm \frac{\sqrt{\left(A\sqrt{\Sigma(0)(\sigma_u^2)^3} + \sigma_u^2 \left(2 + \sqrt{4 + A^2\Sigma(0)\sigma_u^2}\right)\right) \log\left(\frac{\sqrt{2 + A^2\Sigma(0)\sigma_u^2} + A\sqrt{\Sigma(0)\sigma_u^2(4 + A^2\Sigma(0)\sigma_u^2)}}{\sqrt{2}}\right)}}{A\sigma_u^2} \quad (\text{A.75})$$

hence $\Gamma^* = |M^*(0)|$. Since

$$\left.\frac{\partial(\tilde{\Pi}_0 - \Pi_0)}{\partial M(0)}\right|_{M(0)=\Gamma^*} < 0$$

then we can conclude that for any $\Gamma > \Gamma^*$, $\Pi_0 > \tilde{\Pi}_0$. This and the sign of the partial derivatives of Γ^* with respect to $(\Sigma(0), \sigma_u^2, A)$ are obtained through straightforward but computationally intense algebra, that we prefer to omit. $\square$

¹⁷If $w \sim \mathcal{N}(0, \Sigma)$, then

$$\mathbb{E}[\exp(w'Aw + b'w + c)] = |I - 2\Sigma A|^{\frac{1}{2}} \exp\left[\frac{1}{2}b'(I - 2\Sigma A)^{-1}\Sigma b + c\right]$$

Appendix B: Transparent Market Equilibrium in Discrete-time

Let $v \sim \mathcal{N}(p_0, \Sigma_0)$. The informed trader has perfect information in the sense she observes at t_0 the liquidation value v . Let Δx_n be the order placed by the monopolistic insider trader at the n -th auction. The aggregate order at the n -th auction by liquidity traders, who trade for liquidity reasons, is denoted by Δu_n . We assume $\Delta u_n \sim \mathcal{N}(0, \sigma_u^2 \Delta t_n) \forall n$, serially uncorrelated and independent of v . Furthermore, let Δy_n be the total order flow observed by the market maker in period n : $\Delta y_n = \Delta x_n + \Delta u_n$. As in Holden and Subrahmanyam (1994) we denote with W_0 the initial wealth of the informed trader and with W_n the wealth at auction n . Moreover, the informed trader has negative exponential utility, with risk aversion coefficient A , for terminal wealth (denoted by W_{N+1}):

$$U(W_{N+1}) = -\exp\{-AW_{N+1}\} \quad (\text{B.1})$$

The information of the market maker is represented by a *prior* (pre-disclosure) and an *updating* (post-disclosure) filtrations, denoted respectively by $\mathbb{F}$ and $\mathbb{F}'$, whereas the information of the informed agent is represented by a single filtration $\mathbb{G}$:

$$\begin{aligned} \mathbb{F} &\equiv \{\mathcal{F}_n | 0 \leq n \leq N\} & \text{where} & \quad \mathcal{F}_n = \sigma(y_m : 0 \leq m \leq n) \vee \sigma(x_m : 0 \leq m < n) \\ \mathbb{F}' &\equiv \{\mathcal{F}'_n | 0 \leq n \leq N\} & \text{where} & \quad \mathcal{F}'_n = \sigma(y_m, x_m : 0 \leq m \leq n) \\ \mathbb{G} &\equiv \{\mathcal{G}_n | 0 \leq n \leq N\} & \text{where} & \quad \mathcal{G}_n = \sigma(v) \vee \sigma(p_m, p_m^* : 0 \leq m < n) \end{aligned}$$

The monopolistic insider behaves strategically by choosing a $\mathcal{G}$ -measurable function as optimal trading strategy that maximizes her expected utility conditional on all public and private information. Let $J(W_n)$ denote the indirect utility from W_n .

Let $\Delta x_n = X_n(\mathcal{G}_n)$ and $p_n = P_n(\mathcal{F}_n)$ represent the optimal strategies of the insider trader and of the market maker at the n -th auction, respectively. Then, let us define $X = \langle X_1, \dots, X_N \rangle$ and $P = \langle P_1, \dots, P_N \rangle$ as the vectors of strategy functions. The equilibrium concept is defined below.

Definition 2. *An equilibrium of the discrete-time trading game is defined as a triple (X, P, P^*) such that the following conditions hold:*

- i. *For any $n \in \{1, \dots, N\}$ and for any alternative trading strategy $X' = \langle X'_1, \dots, X'_N \rangle$:*

$$\mathbb{E}\left[J\left(W_{n+1}(X, P, P^*)\right) \middle| \mathcal{G}_n\right] \geq \mathbb{E}\left[J\left(W_{n+1}(X', P, P^*)\right) \middle| \mathcal{G}_n\right] \quad (\text{B.2})$$

- ii. *For any $n \in \{1, \dots, N\}$ the prices are competitive:*

$$p_n = \mathbb{E}[v | \mathcal{F}_n] \quad (\text{B.3})$$

Given the definition of the equilibrium we restrict our attention on the class of linear equilibria. In particular, we search for an equilibrium in which both the insider trader and the market maker's strategies are linear functions. Moreover, without loss of generality we consider auctions occurring at equally spaced intervals $\Delta t_n = \Delta t = 1/N$ for all n .

Theorem 3. *There exists a recursive linear equilibrium in which the constants β_n , λ_n , γ_n , α_n , η_n , $\sigma_{z_n}^2$, and Σ_n characterize the following:*

$$\Delta x_n = \beta_n M_{n-1} \Delta t + \Delta z_n \quad (\text{B.4})$$

$$\Delta z_n \sim \mathcal{N}(0, \sigma_{z_n}^2 \Delta t) \quad (\text{B.5})$$

$$p_n = p_{n-1}^* + \lambda_n \Delta y_n \quad (\text{B.6})$$

$$p_n^* = p_{n-1}^* + \gamma_n \Delta x_n \quad (\text{B.7})$$

$$\Sigma_n = \mathbb{V}\text{ar}[v | \mathcal{F}_n'] \quad (\text{B.8})$$

$$J(W_n) = -\eta_{n-1} \exp \left\{ -A \left(W_n + \alpha_{n-1} (v - p_{n-1}^*)^2 \right) \right\} \quad (\text{B.9})$$

$$W_n = W_{n-1} + \Delta x_{n-1} (v - p_{n-1}) \quad (\text{B.10})$$

for all auctions $n \in \{1, \dots, N\}$, where $M_{n-1} \equiv v - p_{n-1}^*$. Given the initial condition $p_0^* = p_0$, and denoting with λ the positive root of the following polynomial, that satisfy the SOC of the insider maximization problem at the last trading round,

$$\lambda^4 \left[A^2 \sigma_u^2 (\sigma_u^2 \Delta t)^2 \right] + \lambda^3 \left[4A \sigma_u^2 (\sigma_u^2 \Delta t) \right] + \lambda^2 \left[4\sigma_u^2 \right] - \lambda \left[A \sigma_u^2 \Delta t \Sigma_0 \right] - \Sigma_0 = 0 \quad (\text{B.11})$$

the constants λ_n , γ_n , α_n , η_n , β_n , $\sigma_{z_n}^2$, Σ_n are given by

$$\lambda_n = \lambda \quad (\text{B.12})$$

$$\gamma_n = \gamma = 2\lambda + A\lambda^2 \sigma_u^2 \Delta t \quad (\text{B.13})$$

$$\eta_n = 1 \quad (\text{B.14})$$

$$\beta_n = \left(\frac{\Lambda \sigma_u^2}{\Sigma_0} \right) \frac{N}{N - n + 1} \quad (\text{B.15})$$

$$\sigma_{z_n}^2 = \sigma_u^2 \left(\frac{\Lambda}{\gamma} \right) \frac{N - n}{N - n + 1} \quad (\text{B.16})$$

$$\Sigma_n = \Sigma_0 \frac{N - n}{N} \quad (\text{B.17})$$

for all auctions $n \in \{1, \dots, N\}$, and by

$$\alpha_n = (2\gamma)^{-1} \quad (\text{B.18})$$

$$\alpha_N = 0 \quad (\text{B.19})$$

for all auctions $n \in \{1, \dots, N - 1\}$, where the constant Λ is defined as follows:

$$\Lambda = \gamma(1 + A\lambda \sigma_u^2 \Delta t)^{-1} \quad (\text{B.20})$$

Proof. We proceed by backward induction. Suppose that for constants α_n and η_n

$$J(W_{n+1}) = -\eta_n \exp \left[-A (W_{n+1} + \alpha_n M_n^2) \right] \quad (\text{B.21})$$

We have then

$$\begin{aligned} J(W_n) &= \max_{\Delta x} \mathbb{E}[J(W_{n+1})|\mathcal{G}_n] \\ &= \max_{\Delta x} \mathbb{E}_n[-\eta_n \exp[-A(W_n + \Delta x(v - p_n) + \alpha_n M_n^2)]] \end{aligned} \quad (\text{B.22})$$

$$\text{s.t.} \begin{cases} p_n &= p_{n-1}^* + \lambda_n(\Delta x + \Delta u_n) \\ p_n^* &= p_{n-1}^* + \gamma_n \Delta x \end{cases} \quad (\text{B.23})$$

where Δx denotes the control quantity of the insider trader at the n -th auction, and the two constraints highlight the linear functional form of the conjectured pricing functions. Substituting Equations (B.23) in (B.22), which clearly shows the strategic behavior of the insider, and evaluating the conditional expectation for log-normal distributions, we obtain

$$J(W_n) = \max_{\Delta x} -\eta_n \exp \left\{ -A \left[W_n + \Delta x [1 - 2\alpha_n \gamma_n] M_{n-1} + (\Delta x)^2 \left[\alpha_n \gamma_n^2 - \lambda_n - \frac{1}{2} A \lambda_n^2 \sigma_u^2 \Delta t \right] + \alpha_n M_{n-1}^2 \right] \right\} \quad (\text{B.24})$$

The first-order condition (FOC), where Δx_n denote the optimized value of Δx for the above expression, turns out to be

$$M_{n-1} [1 - 2\alpha_n \gamma_n] + \Delta x_n [2\alpha_n \gamma_n^2 - 2\lambda_n - A \lambda_n^2 \sigma_u^2 \Delta t] = 0 \quad (\text{B.25})$$

Since the conjectured trading strategy of the informed agent incorporates a random component (the dissimulation part of the trade), it means that in equilibrium the insider must be indifferent across all possible values of her trade, that is the FOC must hold for any Δx_n . This implies that the following conditions (hereafter called *dissimulation conditions*) must hold for any $n \in \{1, \dots, N-1\}$:

$$1 - 2\alpha_n \gamma_n = 0 \quad (\text{B.26})$$

$$\alpha_n \gamma_n^2 - \lambda_n - \frac{1}{2} A \lambda_n^2 \sigma_u^2 \Delta t = 0 \quad (\text{B.27})$$

Note that, if these conditions are satisfied, both FOC and SOC hold.

In equilibrium the conjectured indirect utility must be correct. This means that α_{n-1} and η_{n-1} must be such that Equation (B.21) at time t_n is equal to Equation (B.22):

$$-\eta_{n-1} \exp[-A(W_n + \alpha_{n-1} M_{n-1}^2)] = \max_{\Delta x} \mathbb{E}[J(W_{n+1})|\mathcal{G}_n] \quad (\text{B.28})$$

When computing the expectation in the r.h.s. of the above expression we must take into account that dissimulation of insider trading implies randomness in the insider trading strategy. In particular, $J(W_{n+1})$ contains linear and quadratic functions of two independent (normally distributed) random components: Δu_n and Δz_n . For this purpose let us consider the following lemma.

Lemma 2. *Let X and Z be two independent normally distributed random variables: $X \sim \mathcal{N}(0, \sigma_X^2)$, $Z \sim \mathcal{N}(0, \sigma_Z^2)$. Then, if $\sigma_X^2(2a + c^2 \sigma_Z^2) < 1$:*

$$\mathbb{E}[\exp(aX^2 + bX + dZ + cXZ)] = \exp\left(\frac{\Xi}{2\Phi}\right) \frac{1}{\sqrt{\Phi}} \quad (\text{B.29})$$

where

$$\Xi \equiv d^2\sigma_Z^2 + \sigma_X^2(b^2 + 2d\sigma_Z^2(bc - ad)) \quad (\text{B.30})$$

$$\Phi \equiv 1 - \sigma_X^2(2a + c^2\sigma_Z^2) \quad (\text{B.31})$$

If $\sigma_X^2(2a + c^2\sigma_Z^2) > 1$, the above expectation is not well-defined.

Proof. See Brunnermeier (2001, pag. 64). □

Simple algebra leads to the following expression for the indirect utility at t_n :

$$J(W_n) = -\eta_n \left[\exp(-AQ_n) \mathbb{E}_n \left[\exp \left(a_n(\Delta z_n)^2 + b_n\Delta z_n + d_n\Delta u_n + c_n\Delta z_n\Delta u_n \right) \right] \right] \quad (\text{B.32})$$

where

$$\begin{cases} Q_n & \equiv W_n + [\beta_n(1 - 2\alpha_n\gamma_n) + \beta_n^2(\alpha_n\gamma_n^2 - \lambda_n) + \alpha_n]M_{n-1}^2 \\ a_n & \equiv -A(\alpha_n\gamma_n^2 - \lambda_n) \\ b_n & \equiv -A[(1 - 2\alpha_n\gamma_n) + 2\beta_n(\alpha_n\gamma_n^2 - \lambda_n)]M_{n-1} \\ d_n & \equiv A\lambda_n\beta_nM_{n-1} \\ c_n & \equiv A\lambda_n \end{cases} \quad (\text{B.33})$$

Using Lemma 2 and the two *dissimulation conditions* in Equations (B.26) and (B.27), we find that α and η are constant over time until the second last auction. Therefore, since $\forall n \in \{1, \dots, N-1\}$

$$\alpha_n = [2\lambda_n(2 + A\lambda_n\sigma_u^2\Delta t)]^{-1} \quad \text{and} \quad \alpha_{n-1} = \alpha_n \quad (\text{B.34})$$

it follows that

$$\gamma_{n-1} = \gamma_n \quad (\text{B.35})$$

$$2(\lambda_n - \lambda_{n-1}) + A\sigma_u^2\Delta t(\lambda_n^2 - \lambda_{n-1}^2) = 0 \quad (\text{B.36})$$

which implies

$$\lambda_{n-1} = \lambda_n \quad (\text{B.37})$$

Solving the maximization problem that the insider faces at the last trading round, in which it is clearly optimal not to add any noise component to the order flow, we get as in Holden and Subrahmanyam (1994) the following equations:

$$\beta_N\Delta t = [\lambda_N(2 + A\lambda_N\sigma_u^2\Delta t)]^{-1} \quad (\text{B.38})$$

$$\alpha_{N-1} = [2\lambda_N(2 + A\lambda_N\sigma_u^2\Delta t)]^{-1} \quad (\text{B.39})$$

$$\eta_{N-1} = 1 \quad (\text{B.40})$$

Finally, Equation (B.12) follows from Equation (B.34) evaluated at second last auction and from Equa-

tion (B.39). As in Huddart et al. (2001) market depth is constant over all trading rounds. Moreover, since no utility can be gained after trading is complete, $\alpha_N = 0$ and $\eta_N = 1$.

Market efficiency conditions requires that $p_n = \mathbb{E}[v|\mathcal{F}_n]$ and $p_n^* = \mathbb{E}[v|\mathcal{F}'_n]$. Simple application of the projection theorem for normally distributed random variables confirms the linear pricing rules specified in Equations (B.6) and (B.7), where the slope coefficients are respectively given

$$\lambda_n = \frac{\beta_n \Sigma_{n-1}}{\beta_n^2 \Delta t \Sigma_{n-1} + \sigma_{z_n}^2 + \sigma_u^2} \quad (\text{B.41})$$

$$\gamma_n = \frac{\beta_n \Sigma_{n-1}}{\beta_n^2 \Delta t \Sigma_{n-1} + \sigma_{z_n}^2} \quad (\text{B.42})$$

By the same theorem and using Equation (B.42) we find the expression for the conditional variance:

$$\Sigma_n = \Sigma_{n-1}(1 - \gamma_n \beta_n \Delta t) \quad (\text{B.43})$$

Furthermore, simple algebra allows us to conclude that also γ_n is constant for any auction $n \in \{1, \dots, N\}$.

Now, combining Equations (B.41) and (B.42) we can derive a convenient expression for β_n :

$$\beta_n = \Lambda \sigma_u^2 (\Sigma_{n-1})^{-1} \quad (\text{B.44})$$

where Λ is a constant defined in Equation (B.20). Substituting β_n in Equation (B.43) we clearly see that the conditional variance decreases at a constant rate:

$$\Sigma_n = \Sigma_{n-1} - (\gamma \Lambda \sigma_u^2 \Delta t) \quad (\text{B.45})$$

$$= \Sigma_0 - (\gamma \Lambda \sigma_u^2 \Delta t) n \quad \forall n \in \{1, \dots, N\} \quad (\text{B.46})$$

Since all the private information is incorporated at the end the trading game, $\Sigma_N = 0$. Therefore,

$$0 = \Sigma_0 - (\gamma \Lambda \sigma_u^2 \Delta t) N \quad (\text{B.47})$$

$$\Sigma_n = \Sigma_0 (N - n) / N \quad \forall n \in \{1, \dots, N\} \quad (\text{B.48})$$

Equations (B.15) and (B.16) are obtained through straightforward algebra.

Finally, in order to obtain the optimal constant λ we need to solve the polynomial of degree four in Equation (B.11), obtained by combining Equations (B.38), (B.39) and (B.48). Note that any positive root satisfies the SOC of the insider maximization problem at the last trading round: $-2\lambda - A\lambda^2 \sigma_u^2 \Delta t \leq 0$. This completes the proof of the theorem. Note that if we set $A = 0$ we obtain the same equilibrium as in Huddart et al. (2001), in which case the positive root of the polynomial in Equation (B.11) is equal to $\sqrt{\Sigma_0 / 4\sigma_u^2}$. $\square$

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TABLE 1. EQUILIBRIA

The three graphs in each Panel contrast over time the conditional variance of prices (an inverse measure of market efficiency), $\Sigma(t)$, the sensitivity of the pricing function to the total order flow (an inverse measure of market liquidity), $\lambda(t)$, and the sensitivity of the insider's trading strategy to private information (a direct measure of insider aggressiveness) $\beta(t)$, respectively for the continuous-time equilibrium (i) in a transparent market (solid line), and (ii) in an opaque market (dashed line). Panel A represents the case of risk neutrality ($A = 0$); Panel B the case of risk aversion ($A = 4$). Exogenous parameters are normalized by setting $\Sigma(0) = 1$, $\sigma_u^2 = 1$.

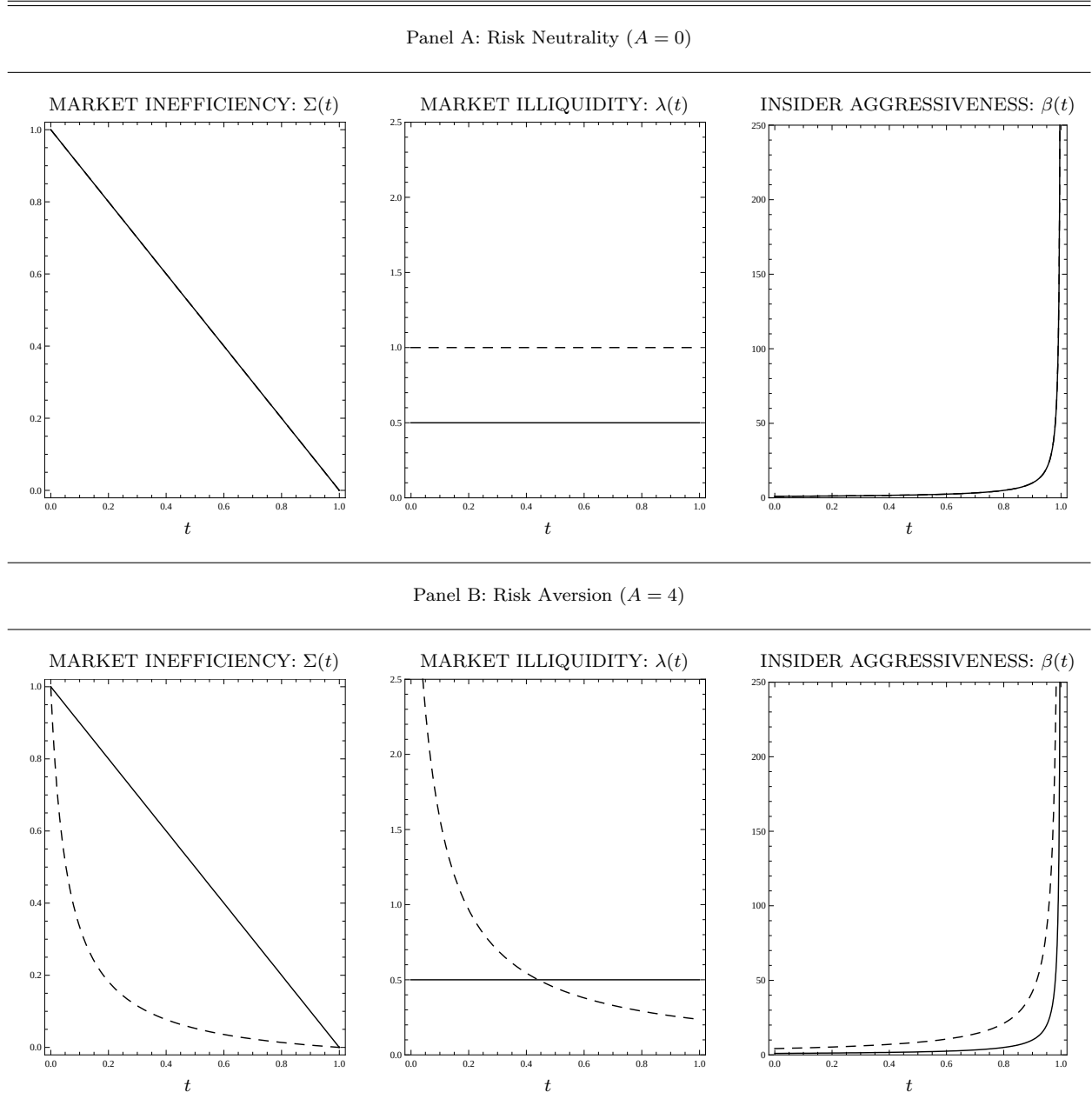
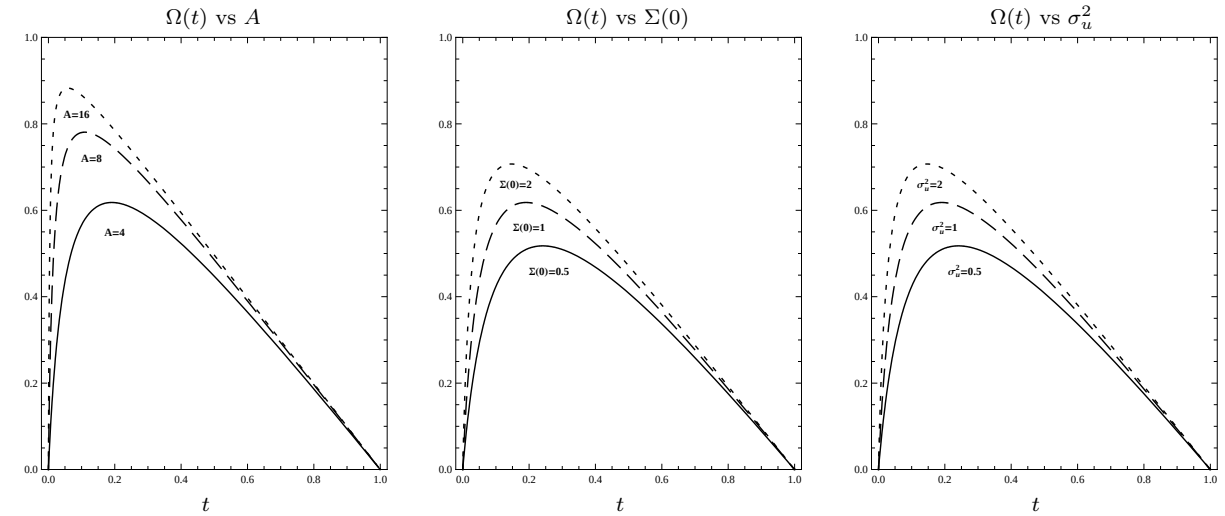


TABLE 2. EFFICIENCY LOSS

The three graphs in Panel A contrast over time the instantaneous efficiency loss and present comparative statics for the exogenous parameters $A, \Sigma(0), \sigma_u^2$. The shaded area in the first graph in Panel B identifies the aggregate measure of efficiency loss (to be scaled by $1/2$). The second and the third graphs in Panel B contrast over the level of risk aversion the aggregate efficiency loss and present comparative statics for the exogenous parameters $\Sigma(0), \sigma_u^2$. Unless otherwise stated, exogenous parameters are normalized by setting $A = 4$, $\Sigma(0) = 1$, and $\sigma_u^2 = 1$.

Panel A: INSTANTANEOUS EFFICIENCY LOSS, $\Omega(t) = (\Sigma(t) - \tilde{\Sigma}(t))/\Sigma(0)$



Panel B: AGGREGATE EFFICIENCY LOSS, $\Omega^a = 2 \int_0^1 \Omega(t) dt$

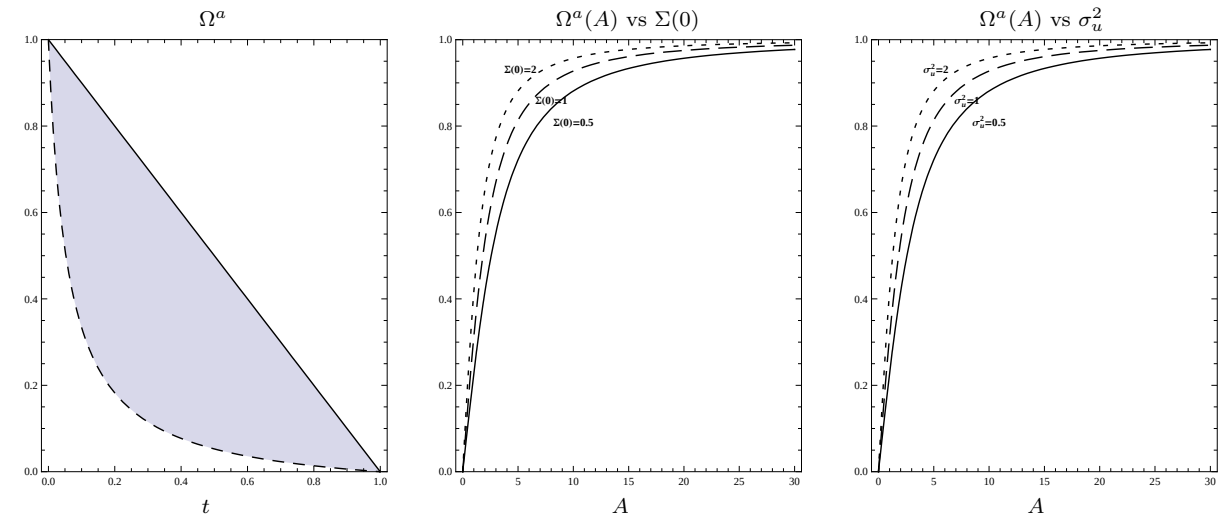
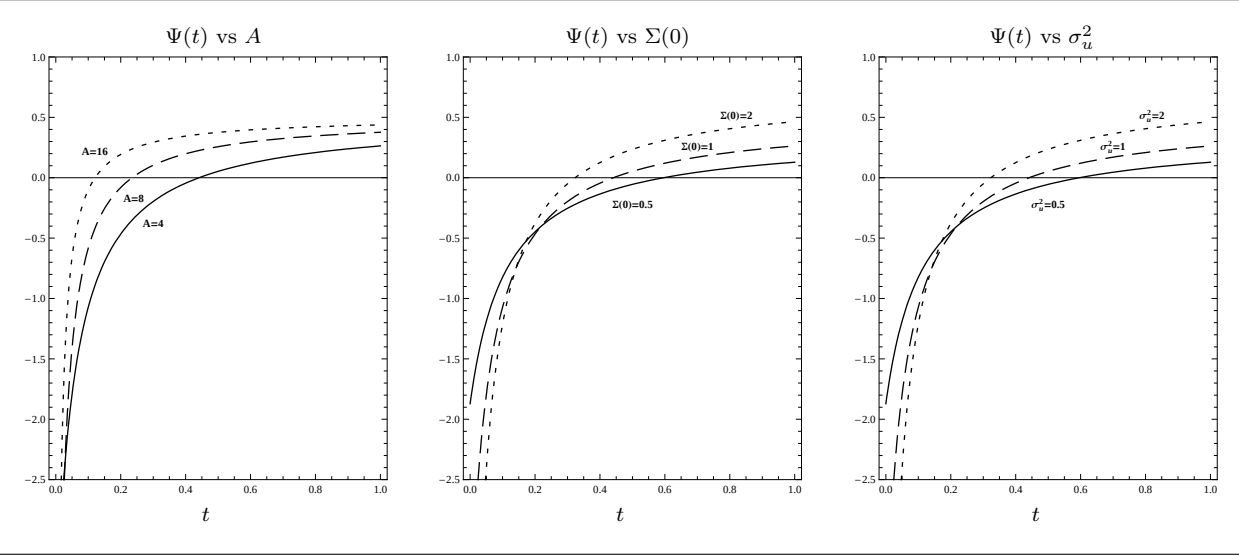


TABLE 3. LIQUIDITY LOSS

The three graphs in Panel A contrast over time the instantaneous liquidity loss and present comparative statics for the exogenous parameters $A, \Sigma(0), \sigma_u^2$. The shaded area in the first graph in Panel B identifies the aggregate measure of liquidity loss. The second and the third graphs in Panel B contrast over the level of risk aversion the aggregate liquidity loss and present comparative statics for the exogenous parameters $\Sigma(0), \sigma_u^2$. Unless otherwise stated, exogenous parameters are normalized by setting $A = 4$, $\Sigma(0) = 1$, and $\sigma_u^2 = 1$.

Panel A: INSTANTANEOUS LIQUIDITY LOSS, $\Psi(t) = (\lambda(t) - \bar{\lambda}(t))\sigma_u^2$



Panel B: AGGREGATE LIQUIDITY LOSS, $\Psi^a = \int_0^1 \Psi(t)dt$

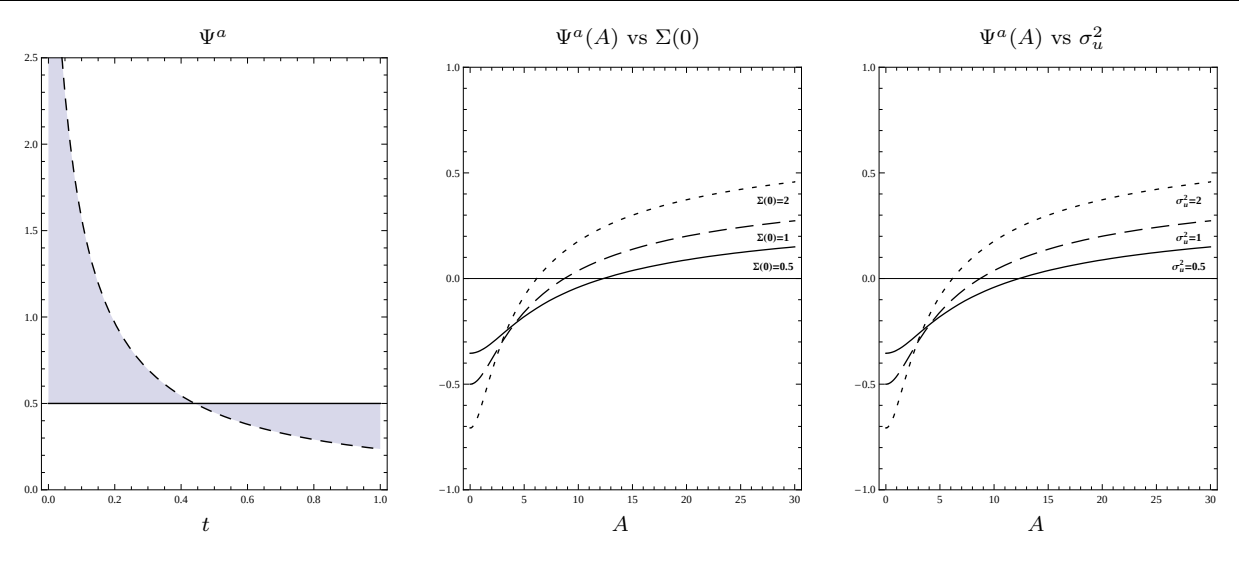
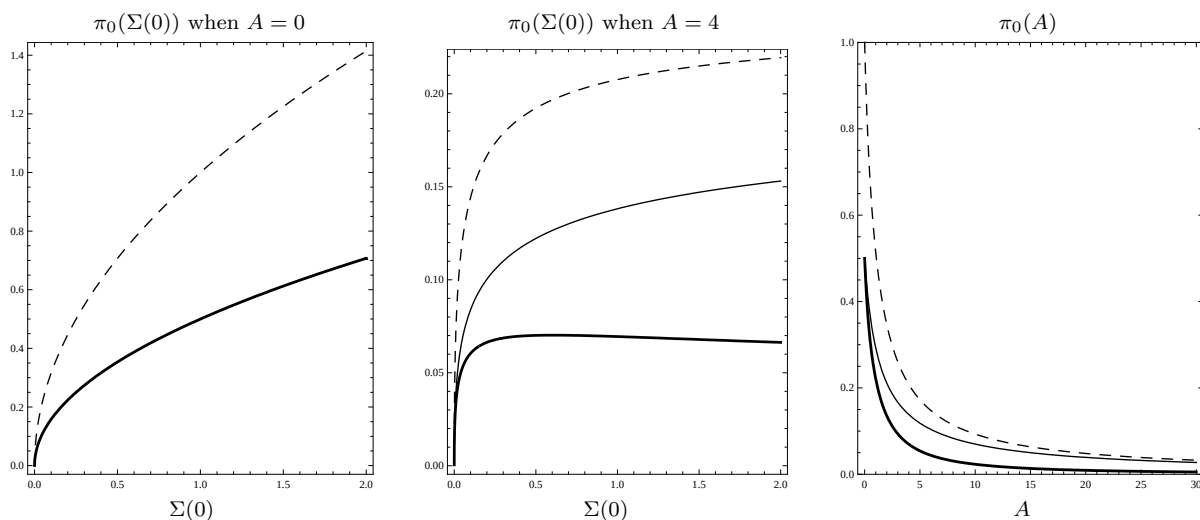


TABLE 4. INSIDER TRADING PROFITABILITY

The three graphs in Panel A contrast the unconditional expected utility of the insider over the prior variance of the private information (the first two graphs) and the level of risk aversion (third graph). The first two graphs in Panel B contrast the conditional expected utility of the insider over her informational advantage, $\Gamma \equiv |v - \mathbb{E}_0(v)|$, while the third graph contrast the minimum level of the insider's informational advantage, Γ^* such that for any $\Gamma > \Gamma^*$ her conditional expected utility is higher in a transparent market. The dashed line represents an opaque market, the solid line a transparent market, and the solid bold line the difference between the two. In the 3-dimensional graph the darker surface represents the threshold level Γ^* as a function of $\Sigma(0)$ and A ; the brighter surface as a function of σ_u^2 and A . Unless otherwise stated, exogenous parameters are normalized by setting $W(0) = 0$, $A = 4$, $\Sigma(0) = 1$, and $\sigma_u^2 = 1$.

Panel A: UNCONDITIONAL EXPECTED UTILITY, $\pi_0 = \mathbb{E}_0[U(W(1))]$



Panel B: CONDITIONAL EXPECTED UTILITY, $\Pi_0 = \mathbb{E}_0[U(W(1))|v]$

